

# Modelling Financial Markets

## Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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# Opening: Can Past Prices and Public Information Predict Returns?



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*"A market in which prices always 'fully reflect' available information is called 'efficient.'"*
— *Fama (1970)*

- Lag-1 autocorrelation of daily returns: $-0.099$ for the S&P 500 and $0.109$ for the BET, both significant
  - Bitcoin: $-0.022$, inside the robust test band, so we do not reject $\rho_1 = 0$
- Yesterday's return explains 0.98% of the variance of today's return for the S&P 500 and 1.19% for the BET
- Dividend–price ratio, 680 monthly forecasts for the S&P 500: out-of-sample $R^2 = -0.51\%$
  - the forecast is worse than the historical mean
- Of 2 000 random trading rules, 5.2% are "significant" at 5%
  - the best one has $|t| = 3.30$, although it is pure chance

▸ Central chart: Lag-1 Autocorrelation Across Markets

## Chapter outline

- Can past prices, news or other public information predict returns after adjusting for risk and costs?
- Sections 1–5 contain the core material and end with a short check.
- The further-study topics contain the derivations, the other tests and the project topics.
- Detailed examples and advanced topics are in Further Topics, after the Synthesis
  - the footer button leads there; the Back button returns to the starting slide


- 1. Returns and information
- 2. Autocorrelation and variance ratios
- 3. Out-of-sample predictability
- 4. Event studies
- 5. Multiple testing and anomalies
- Topics for further study


### What you should be able to do

- derive the random-walk restrictions and the variance ratio
- compute the lag-1 autocorrelation and the variance ratio robustly
- compute the out-of-sample $R^2$ and the Clark–West test
- compute the cumulative abnormal return in an event study
- interpret an anomaly after the Holm correction
- Further Topics and Appendix: reading for familiarity

# 1. Returns and information

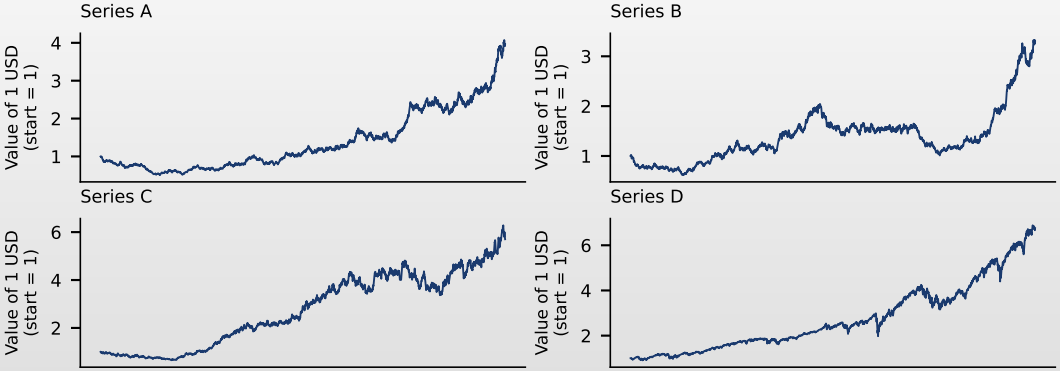
- **RW1**: Random Walk 1 — independent, identically distributed increments.
- **RW2**: Random Walk 2 — independent increments, possibly with different distributions.
- **RW3**: Random Walk 3 — uncorrelated increments, possibly dependent.

- The question: what does it mean, precisely, that past prices contain no information?
  - information about future returns
- The version matters: a test built under RW1 can reject only because volatility varies over time.
- Every test also needs a model of expected returns, so a rejection does not by itself prove inefficiency.

# A Game: Which Series Is the Real Market? (1/2)

**S&P 500**: Standard & Poor's 500.

One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility




MFM_ch2_random_walk_game ▶ Interactive

# A Game: Which Series Is the Real Market? (2/2)

- Which panel is the S&P 500? Vote: A, B, C or D?

MFM_ch2_random_walk_game

# A Game: Which Series Is the Real Market? (2/2)

- Which panel is the S&P 500? Vote: A, B, C or D?
- Answer: **D**.
- Roberts (1959): simulated random walks produce "patterns" by chance
- We need statistical tests, not visual inspection
- Bachelier (1900) and Samuelson (1965): properly anticipated prices fluctuate randomly
  - so unpredictability is the mark of a well-functioning market

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# Efficiency and the Information Set

- Fama (1970): a market is **efficient** if prices "fully reflect" all available information $\mathcal{I}_t$
  - Efficient Market Hypothesis, EMH
  - portrait and surveys: ▸ Appendix
  - precursors: ▸ Appendix
  - formally: no strategy using $\mathcal{I}_t$ earns **expected** returns above the fair (risk-adjusted) return

| Form | Information set $\mathcal{I}_t$ | Typical test (this chapter) |
|---|---|---|
| Weak | Past prices and returns | Autocorrelation, variance ratio; in the appendix: runs and long-memory tests |
| Semi-strong | All public information (news, reports) | Event studies |
| Strong | Public and private (insider) information | Returns of insiders and fund managers |

- The strong form implies the semi-strong form, which implies the weak form
- Risk-adjusted returns with factor models follow in Chapter 3

## The Joint-Hypothesis Problem

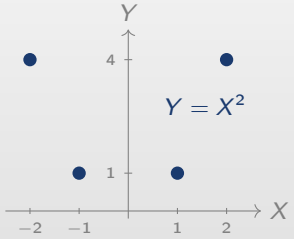
**CAPM**: Capital Asset Pricing Model.

- "Fair return" requires a model of expected returns
  - e.g. CAPM, Chapter 3
- Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model Fama (1991)
  - A rejection may mean an inefficient market **or** a wrong model of risk
  - Predictable **expected** returns are allowed if they compensate time-varying risk
- At daily horizons expected returns are tiny
  - so the choice of model matters little
  - weak-form tests are the cleanest
- Efficiency can never be "proved", only not rejected by a given test on given data

## Correlation and dependence in returns

- "The past contains no information" can mean zero correlation or independence: the two differ.
- Lag-k autocorrelation is the correlation between a return and the return k periods earlier.
- Zero correlation excludes linear association, but may leave dependence in squared returns.

- Toy example: $X = -2, -1, 1, 2$, equally likely; $Y = X^2$.
- $\mathbb{E}[X] = 0$ and $\mathbb{E}[XY] = (-8 - 1 + 1 + 8)/4 = 0$, so the correlation is zero.
- Yet knowing $X$ determines $Y$ exactly: they are dependent.



## The Random Walk: the Model

$$p_t = \mu + p_{t-1} + \varepsilon_t, \qquad p_t = \ln P_t$$

- $P_t$: the index level at the close of day $t$.
  - $p_t$: the log price.
  - $\mu$: the drift (mean growth per day).
  - $\varepsilon_t$: the day's shock, with mean zero and variance $\sigma^2$.
- By repeated substitution: $p_t = p_0 + \mu t + \sum_{s=1}^{t} \varepsilon_s$.
  - the log price is a linear trend plus the sum of all past shocks.
- Equivalently, the log return $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$.
- The best forecast: $\mathbb{E}[p_{t+h} \mid \mathcal{I}_t] = p_t + \mu h$, with $\mathcal{I}_t = \{p_t, p_{t-1}, \dots\}$.
  - it holds if the shocks have conditional mean zero, $\mathbb{E}[\varepsilon_{t+1} \mid \mathcal{I}_t] = 0$: today's price plus the drift.
  - RW3 (zero autocorrelations only) is not enough.
  - Under RW3, the past does not, in general, improve the best *linear* forecast.
- With uncorrelated shocks of variance $\sigma^2$: $\mathrm{Var}(p_{t+h} - p_t) = h\sigma^2$.
  - the variance grows linearly with the horizon.
  - the basis of the variance ratio in section 2.
- RW1, RW2 and RW3 differ only in what they assume about the shocks $\varepsilon_t$ Campbell, Lo & MacKinlay (1997).

# RW1, RW2, RW3: the restrictions on returns

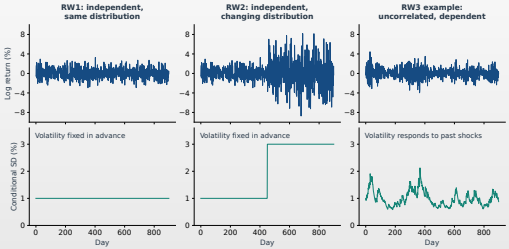
- **i.i.d.**: independent and identically distributed.
- **GARCH**: Generalized AutoRegressive Conditional Heteroskedasticity.
- $z_t$: a shock with the standard Normal distribution $N(0,1)$.
  - in GARCH(1,1), $\omega > 0$ and $\alpha, \beta \geq 0$.

| | Condition on the shocks | Illustrative return model |
|---|---|---|
| RW1 | $\varepsilon_t$ i.i.d., mean 0, variance $\sigma^2$ | $r_t = \mu + 0.01 z_t$, $z_t$ i.i.d. $N(0,1)$ |
| RW2 | $\varepsilon_t$ independent; the variance $\sigma_t^2$ may vary over time | $r_t = \mu + \sigma_t z_t$, $\sigma_t = 1\%$ then 3%, fixed in advance |
| RW3 | $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for every $k \geq 1$ | GARCH: $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ |

- RW1 implies RW2: i.i.d. returns are in particular independent.
- RW2 implies RW3: independent returns are in particular uncorrelated, provided their variance exists.
- A GARCH model keeps zero correlation but violates RW2.
  - the size of today's move depends on past shocks.
- RW3 does not make tomorrow's mean unpredictable: a nonlinear function of the past can forecast it.

# Three models for returns and volatility

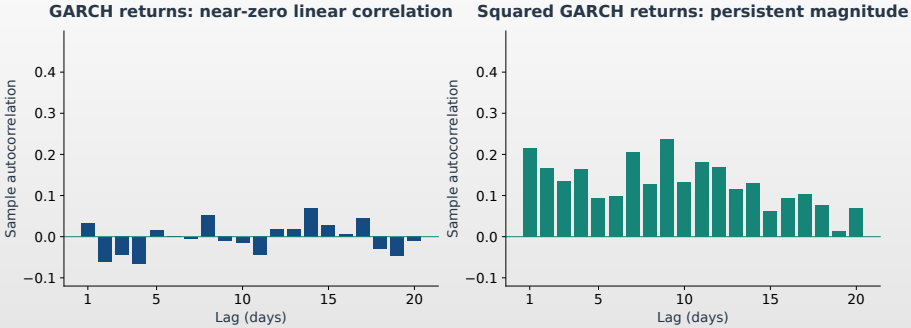
**SD**: Standard Deviation.




- Three simulated series with mean zero.
  - daily returns on top; below, the conditional standard deviation used in the simulation.
- RW2 volatility is fixed in advance, while GARCH volatility responds to past shocks.
  - 1% then 3%, from the middle of the sample.
- Volatility clustering: large moves tend to be followed by large moves.
- The RW2 series also has a turbulent period although its returns are independent.
  - The chart alone cannot tell a regime change fixed in advance from volatility clustering.

▶ Interactive

# GARCH: returns versus squared returns




- Left: the autocorrelations of simulated GARCH returns are close to zero at all lags.
- Right: the autocorrelations of squared returns are positive up to lag 20.
  - a day with a large move is followed by other days with large moves.

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## Check: returns and information

- Can returns remain independent if their standard deviation moves from 1% to 3% on a date fixed in advance?
- Does zero autocorrelation establish independence?
- Why does rejecting an efficiency test not, on its own, prove that the market is inefficient?

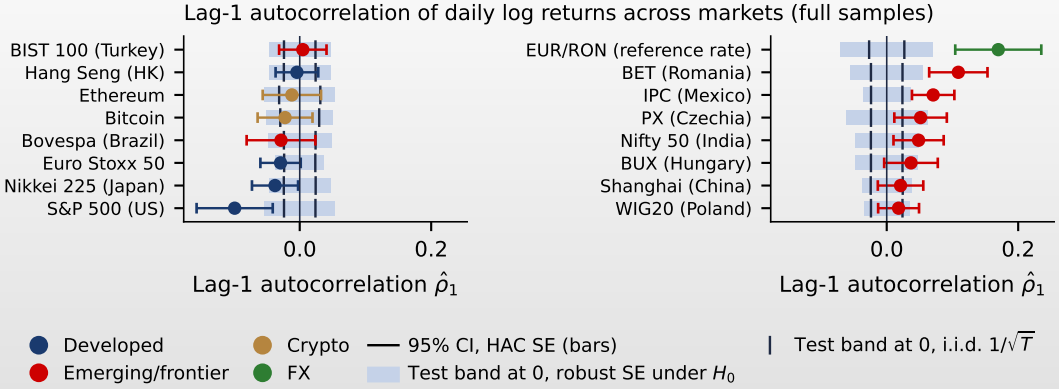
## Answers: returns and information

- Yes. The series satisfies RW2 but violates RW1, because the distribution is no longer identical.
- No. In the GARCH example, return autocorrelation is zero, but squared returns are dependent.
- Every efficiency test also tests the model of expected returns.
- A rejection can come from a wrong risk model (the joint-hypothesis problem).

## 2. Autocorrelation and variance ratios

- The question: does today's return predict tomorrow's in equity, crypto and currency markets?
- The lag-1 autocorrelation $\rho_1$ measures linear dependence between consecutive days.
  - The variance ratio accumulates the autocorrelations up to horizon $q$.
- Markets: 13 stock indices, Bitcoin, Ethereum and the EUR/RON reference rate.
  - developed and emerging, including the BET.
- Why this selection: predictability is more likely:
  - where liquidity is thin (BET, crypto).
  - or where the exchange rate is managed (EUR/RON).
- Samples: all to 18 September 2026.
  - equities from January 2000, EUR/RON from July 2005.
  - Bitcoin from September 2014 and Ethereum from January 2016.

# Lag-1 Autocorrelation Across Markets (1/3)




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## Lag-1 Autocorrelation Across Markets (2/3)

**SE**: Standard Error; **HAC**: Heteroskedasticity and Autocorrelation Consistent; **CI**: Confidence Interval.

- Each dot is the lag-1 autocorrelation $\hat{\rho}_1$ of one market's daily log returns, over its full sample.
- The coloured bar is the 95% confidence interval $\hat{\rho}_1 \pm 1.96\,\mathrm{SE}_{\mathrm{HAC}}$.
  - $\mathrm{SE}_{\mathrm{HAC}}$: the standard error of the estimator, valid also when $\rho_1 \neq 0$ (formula).
- The light band is the robust test band $0 \pm 1.96\,\mathrm{SE}_0$; the ticks mark the i.i.d. band $0 \pm 1.96/\sqrt{T}$.
  - $\mathrm{SE}_0$: derived under $H_0$ (unpredictable mean), test construction; $T$: the number of daily returns.
- Reading rule: we reject $\rho_1 = 0$ at 5% if the dot lies outside the robust test band.
  - Nikkei 225 and PX: the HAC interval excludes 0, but $\hat{\rho}_1$ stays inside the test band, so we do not reject.
- In all 16 series, $\mathrm{SE}_0$ exceeds the i.i.d. value; for the S&P 500 ($T = 6\,717$) it is 0.027 against 0.012.
- BIST, WIG20, BUX, PX, IPC and BET: stock-exchange indices.
  - of Istanbul, Warsaw, Budapest, Prague, Mexico and Bucharest.
  - FX is foreign exchange (the EUR/RON reference rate).

## Lag-1 Autocorrelation Across Markets (3/3)

- BET ($\hat{\rho}_1 = 0.109$), IPC (0.071), Nifty 50 (0.048) and EUR/RON (0.170): positive and significant autocorrelations.
- S&P 500: $\hat{\rho}_1 = -0.099$, negative and significant.
- Bitcoin ($\hat{\rho}_1 = -0.022$) and Ethereum ($-0.012$): inside the robust test band, so we do not reject $\rho_1 = 0$.
- For the BET, persistence can come from thin trading in some stocks, whose prices absorb news with a delay.
- For EUR/RON, it can come from the managed float, in which the central bank smooths exchange-rate moves.
- For the S&P 500, the reversal can come from short-term overreaction and liquidity provision.
- The effect is small: $\hat{\rho}_1^2$ is 0.98% for the S&P 500 and 1.19% for the BET.
  - the share of today's return variance explained by yesterday's return.
- Both robust standard errors require a finite fourth moment, which Chapter 1 questions.
- What the result supports: description.
  - The proposed mechanisms (thin trading, managed float, overreaction) are not tested here.

MFM_ch2_autocorrelation

## The Variance Ratio: Why and What It Measures

**VR**: Variance Ratio.

- Why another test: $\hat{\rho}_1$ looks at a single lag.
  - small autocorrelations at many lags can each pass the test yet accumulate over longer horizons.
- **Variance ratio** over a horizon of $q$ days Lo & MacKinlay (1988): $\mathrm{VR}(q) = \dfrac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}$.
  - equal to 1 without autocorrelation.
  - $q = 5$: one trading week.
- Without autocorrelation, $\widehat{\mathrm{VR}}(q)$ tends to 1 even when volatility varies over time (RW2, RW3).
- For a covariance-stationary series, $\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\rho_k$.
  - a weighted sum of autocorrelations, with weights that fall with the lag $k$.
- $\mathrm{VR}(q) > 1$ signals persistence, $\mathrm{VR}(q) < 1$ reversal (mean reversion).
- For $q = 2$, $\mathrm{VR}(2) = 1 + \rho_1$: on the S&P 500, $1 - 0.099 = 0.901$, equal to $\widehat{\mathrm{VR}}(2)$ estimated directly.

## The robust variance-ratio band

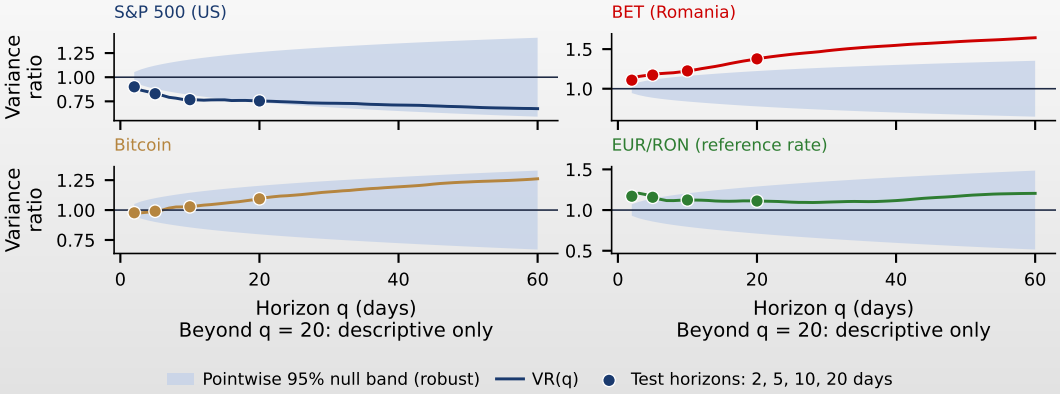
- The robust 95% band is centred at 1, the value under the null.
  - if $\widehat{\mathrm{VR}}(q)$ falls outside it, we reject the null at 5%.
- The band is not a confidence interval.
  - a confidence interval would be centred at the estimate $\widehat{\mathrm{VR}}(q)$, not at 1.
- The robust null band is $1 \pm 1.96\sqrt{\hat{\theta}(q)/T}$ (test construction).
  - $T$: the number of daily returns.
  - $\hat{\theta}(q)$: the estimated variance of $\sqrt{T}\,(\widehat{\mathrm{VR}}(q) - 1)$, robust to heteroskedasticity.
- When large moves follow large moves (volatility clustering), $\hat{\theta}(q)$ grows and the band widens.
- $\hat{\theta}(q)$ also grows with $q$, so the band widens at long horizons.
  - a sample contains fewer independent $q$-day periods.


▸ Appendix: a numerical check

## Which variance-ratio horizons do we test?

- Why several horizons: reversal or persistence can appear at one day, a week or a month.
  - we do not know in advance at which.
- We test four horizons, $q = 2, 5, 10, 20$ trading days.
  - two days, about one week, two weeks and one trading month.
- The choice follows Lo & MacKinlay (1988), who use $q = 2, 4, 8, 16$ on weekly data.
  - other horizons can give a different result.
- The robust null band has level 5% at each horizon separately.
  - with four tests, a false rejection occurs more often than 5% of the time.
- The Chow–Denning test Chow & Denning (1993) corrects this.
  - it uses the largest $|z^*(q)|$ across the four horizons.
  - The critical value is 2.49 instead of 1.96.
- The Chow–Denning test rejects RW3 for 5 of the 16 series, including the BET and the S&P 500.
- For Bitcoin, the Chow–Denning test does not reject RW3 (p-value $= 0.835$).
- $z^*$ and the Chow–Denning test require a finite fourth moment, which Chapter 1 questions.

# Variance-Ratio Profiles (1/2)




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## Variance-Ratio Profiles (2/2)

- Each curve shows the estimated variance ratio $\widehat{\mathrm{VR}}(q)$ for horizons $q$ from 2 to 60 days.
  - $q = 60$, about three trading months, is only descriptive.
- The shaded area is the robust 95% band around 1, the value without autocorrelation.
- S&P 500: $\widehat{\mathrm{VR}}(q)$ is below 1 (reversal) and leaves the band at 2, 5 and 10 days
  - at 20 days it is just inside the band: 0.754 against the edge 0.740
  - $\widehat{\mathrm{VR}}(2) = 0.901$, $\widehat{\mathrm{VR}}(5) = 0.829$, $\widehat{\mathrm{VR}}(10) = 0.767$, $\widehat{\mathrm{VR}}(20) = 0.754$; $\widehat{\mathrm{VR}}(60) = 0.674$
- BET: $\widehat{\mathrm{VR}}(q)$ is above 1 (persistence), rises with $q$ and leaves the band at all four tested horizons
  - $\widehat{\mathrm{VR}}(2) = 1.108$, $\widehat{\mathrm{VR}}(5) = 1.173$, $\widehat{\mathrm{VR}}(10) = 1.225$, $\widehat{\mathrm{VR}}(20) = 1.377$; $\widehat{\mathrm{VR}}(60) = 1.644$
- Bitcoin stays inside the band at all horizons: the null is not rejected.
- EUR/RON: $\widehat{\mathrm{VR}}(q)$ stays above 1 at every horizon.
  - but leaves the band only at 2 and 5 days: $\widehat{\mathrm{VR}}(2) = 1.170$, $\widehat{\mathrm{VR}}(5) = 1.157$.
- What the result supports: description.

MFM_ch2_variance_ratio

## Check: autocorrelation and variance ratios

- Why can an i.i.d. band be misleading under volatility clustering?
- Does positive autocorrelation imply profit after costs?
- What does a variance ratio below one suggest?
- Transfer: a test statistic has magnitude 2.4.
  - If the VR estimate stays the same but its standard error doubles, what does the magnitude become?
- Is the null still rejected at 5% (two-sided)?

## Answers: autocorrelation and variance ratios

- The i.i.d. standard error ignores dependence in squared returns, so it is too small.
  - The test rejects the null too often.
- No: effect size, risk, implementation and costs still matter.
- A variance ratio below 1 suggests reversal over the chosen horizon.
- To call the deviation significant, we need a robust standard error.
- The statistic halves to 1.2.
- No: 1.2 does not cross 1.96.
  - the estimated effect is unchanged, its evidence is weaker.

## 3. Out-of-sample predictability

- The question: does a variable known today forecast next month's market return?
- The classic candidate is *dp*, the log dividend–price ratio.
  - dividends over the last 12 months divided by the average index level in that month.
- In sample, such regressions appear to work.
  - on new data, most published predictors fail Welch & Goyal (2008).
- The proper test: every forecast uses only the data available when it is made.

## How we build the forecasts

**OS**: Out-of-Sample.

- $r$: the monthly S&P 500 log return, with dividends, minus the 3-month US Treasury bill rate.
- Each month $t$ we estimate $r_{s+1} = \alpha + \beta\, dp_s + u_{s+1}$ on all months up to $t$.
  - at the next step the window grows by one month (expanding window).
- The $dp$ forecast for month $t+1$: $\hat{r}_{t+1} = \hat{\alpha} + \hat{\beta}\, dp_t$.
- The competing forecast: the historical mean $\bar{r}_{t+1}$.
  - The mean of returns up to month $t$.
  - it corresponds to no predictability.
- The first forecast is for January 1970, giving 680 monthly forecasts.
- Monthly dividends are interpolated from quarterly totals.
  - so $dp_t$ was not known exactly in month $t$.
  - the exercise is pseudo-out-of-sample.

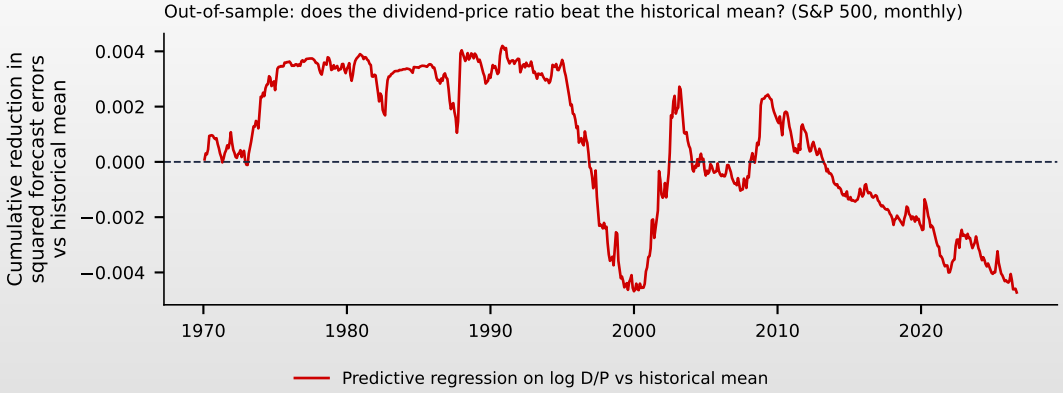
## How we score the result: $R^2_{OS}$

- $R^2_{OS} = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$, with the sums over the forecast months.
- $R^2_{OS} > 0$: the *dp* forecast has smaller squared errors than the historical mean.
- $R^2_{OS} < 0$: the historical mean forecasts better.
  - unlike the in-sample $R^2$, $R^2_{OS}$ can be negative.
- Why the two differ: in sample, $\hat{\beta}$ also fits the noise of the months it explains.
- A positive $R^2_{OS}$ can also arise by chance.
  - we test its significance separately (the Clark–West test, in the results).

## Reading the cumulative squared-error chart

- Each month from 1970 adds $(r_t - \bar{r}_t)^2 - (r_t - \hat{r}_t)^2$.
  - the squared error of the historical mean minus that of the *dp* forecast.
- The curve rises in months in which *dp* forecasts better and falls in months in which it forecasts worse.
- The final value of the curve has the sign of $R^2_{OS}$.
- Vertical axis: the running sum, with returns as fractions (0.01 = 1%), squared.

# The Dividend–Price Ratio: Out-of-Sample Forecasts




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## The Dividend–Price Ratio: Results for the S&P 500 (1/2)

- The curve ends below zero: over 680 monthly forecasts, $R^2_{OS} = -0.51\%$.
- The Clark–West test Clark & West (2007) compares the squared errors and corrects a handicap of the *dp* model.
  - even if $\beta = 0$, estimating $\beta$ adds noise to the forecast.
- On the S&P 500 the test gives $t = 0.42$ (one-sided $p = 0.34$).
  - we do not reject the absence of predictability.
  - the test on squared errors requires a finite fourth moment, which Chapter 1 questions.
- Until 1990 the curve rises: *dp* forecast better than the historical mean.
- In the 1990s boom, *dp* fell to record lows while prices kept rising: the advantage was lost.
- In the 2000s the advantage partly returned, and after 2010 it was lost again.
- What the result supports: comparison of procedures.

## The Dividend–Price Ratio: Results for the S&P 500 (2/2)

- Does a small $R^2_{OS}$ matter economically? Campbell & Thompson (2008) show that it does: even 0.5% is a lot for an investor.
- A mean–variance investor would raise the expected return by about 20%.
  - who raises equity exposure when the forecast is high and cuts it when it is low.
  - formula: Appendix.
- The calculation uses the monthly market Sharpe ratio, 0.160.
- Campbell and Thompson also impose two restrictions.
  - $\hat{\beta}$ with the sign theory predicts.
  - a non-negative equity-premium forecast.
- On the S&P 500 the restrictions bind in 5.7% of months and leave $R^2_{OS}$ at $-0.51\%$.
  - the economic gain is absent.

## Check: out-of-sample predictability

- What does a negative $R^2_{OS}$ mean?
- Why can a regression with a good in-sample $R^2$ fail on new data?
- Can a profitable strategy have a negative $R^2_{OS}$?
- What does the loss of the *dp* advantage in the 1990s say about the stability of the relation?

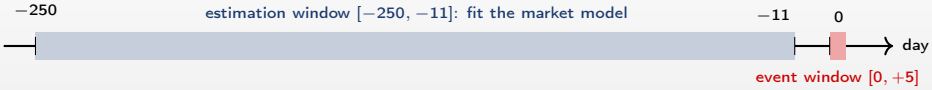
## Answers: out-of-sample predictability

- The historical mean forecast better than the model over the months evaluated.
- In sample, $\hat{\beta}$ also fits the noise.
  - on new data the noise does not repeat.
- Yes: profit can come from constant market exposure or from a few correct sign calls.
  - not from small squared errors.
- The relation is not stable.
  - a predictor that was useful until 1990 loses all its advantage when *dp* falls to record lows.

# 4. Event studies

- The question: how fast and how fully does the price absorb a precisely dated piece of public news?
- To answer it, we need the return the stock would have earned without the news.
- The difference between the observed return and this expected return is the abnormal return (AR).
- An event study measures AR around the news and so tests semi-strong efficiency Fama, Fisher, Jensen & Roll (1969).
- The section's example: the bank tax announced in December 2018.
  - unanticipated news that hit the listed banks directly.

# The Event-Study Method (1/2)




- Market model: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$.
  - $a_i$: the intercept, $b_i$: the market beta of firm $i$.
  - $r_{m,t}$: the market-index return on day $t$.
- The model removes from the firm's return the part explained by the same day's market move.
- With daily data, multi-factor models reduce the variance of $AR$ only slightly MacKinlay (1997).
- Estimation window $[-250, -11]$: about one trading year in which we estimate $a_i$ and $b_i$.
- It stops 10 days before the event, so that any information leak does not enter the estimation.
- Event window $[0, +5]$: the announcement day and five more days.
  - to see whether the reaction continues or reverses.

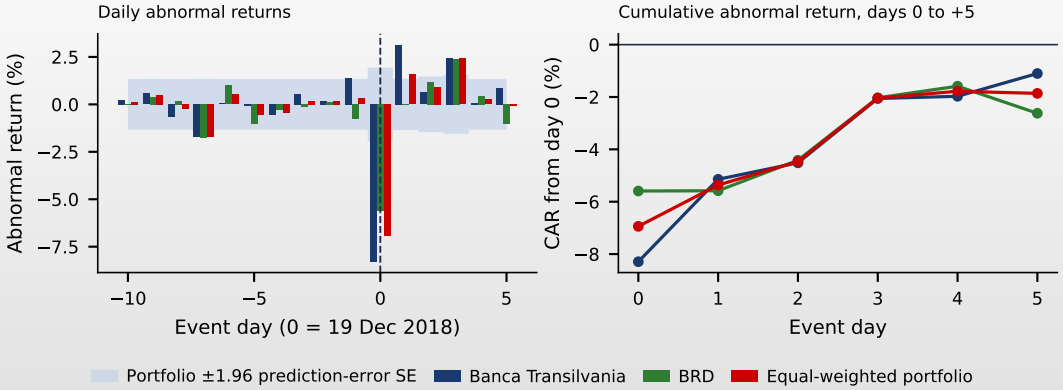
# The Event-Study Method (2/2)

- Abnormal return: $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$, with $\hat{a}_i$, $\hat{b}_i$ estimated in the estimation window.
- Cumulative abnormal return (CAR): $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$, where $[\tau_1, \tau_2]$ are the days summed.
- First-pass test: $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$ MacKinlay (1997).
  - $\hat{\sigma}_\varepsilon$: the standard deviation of the residuals in the estimation window.
  - $L$: the number of days summed.
- The formula ignores the uncertainty in $\hat{a}_i$ and $\hat{b}_i$.
  - its correction appears in the worked example.
- For unanticipated news, efficiency predicts a jump on day 0 and no predictable drift or reversal afterwards
- Daily data can only show absorption within a day.
  - speed at the microsecond scale needs intraday data (Chapter 10).
  - literature: Appendix.
- Application: the price response of Ethereum to network development milestones
  - Quantinar: Market Responses to ETH Development Milestones

## The Bank Tax: the Data and How to Read the Chart

- The government announced the tax on bank assets on the evening of 18 December 2018.
  - event day 0 is 19 December.
- We study Banca Transilvania (TLV), BRD and their equal-weighted portfolio.
  - $r_{m,t}$ is the return of the BET index.
- Left panel: daily $AR$, in percent, from day $-10$ to day $+5$.
- Light-blue band: $\pm 1.96$ standard deviations of the portfolio's prediction error.
  - a portfolio bar outside it is significant at 5%.
- Right panel: $CAR$ from day 0 up to the day on the horizontal axis, in percent.
- A curve that returns towards zero after day 0 indicates a reversed reaction.

# Worked Example: the Romanian Bank Tax of December 2018




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## The Bank Tax of December 2018: Results (1/2)

- For TLV, the estimated market beta is 1.18.
  - the residual standard deviation in the estimation window is 0.83% per day.
- On day 0, the abnormal return of TLV is $AR_0 = -8.3\%$, with $t = -10.0$ by the first-pass formula.
- Also on day 0: $AR_0 = -5.6\%$ for BRD and $-6.9\%$ for the equal-weighted portfolio.
- That day the log return of the BET was $-11.9\%$, a fall in the index of 11.2%.
- With such an extreme market return the uncertainty in $\hat{b}$ matters.
  - the prediction-error standard deviation for TLV rises from 0.83% to 1.21%.
- With this prediction error, $t = -6.85$.
  - the effect remains strongly significant.
- TLV and BRD carry large weights in the BET.
  - so $r_{m,0}$ already contains part of the shock.
  - $AR_0$ understates the fall.

## The Bank Tax of December 2018: Results (2/2)

- Over days $+1, \ldots, +5$, the cumulative abnormal return of TLV is $+7.2\%$ ($t = 3.82$).
- Over days $0, \ldots, +5$, the cumulative abnormal return is only $-1.1\%$ ($t = -0.54$).
  - the abnormal component of TLV's fall, measured against the BET, is almost offset over the next five sessions.
  - this does not show that the share price recovered its loss: over days 0 to $+5$ the log return is $-15.6\%$ for TLV and $-12.7\%$ for the BET benchmark.
- Explanation 1: an **overreaction** on day 0, against semi-strong efficiency.
- Explanation 2: later news about the final form of the ordinance.
- Explanation 3: the benchmark moved with the event.
  - TLV and BRD carry large weights in the BET, which fell by 11.9% (log) on day 0.
  - so *AR* and *CAR* measure TLV against an index that already contains the shock.
  - the BET did not rebound: over days $+1$ to $+5$ its log return is $-0.8\%$.
- A single event does not let us choose between the three explanations.
- What the result supports: causal identification, of the day-0 reaction only.
  - If no other news arrived that day; the later offset stays unexplained.

MFM_ch2_seminar

## Check: event studies

- Why must you specify the event window?
- Could another announcement explain the same movement?

## Answers: event studies

- CAR adds abnormal returns over the days of the window only.
- A window chosen after seeing the data can create or hide an effect.
- Yes: overlapping news is an alternative explanation, so the reaction alone is not causal proof.

## 5. Multiple testing and anomalies

- The question: does a calendar anomaly remain significant?
  - once we account for how many patterns were searched
- The literature has tested thousands of patterns on the same data, and some pass the test by chance alone Harvey, Liu & Zhu (2016).
- A small p-value means something different after 15 tests than after one.
- First we measure how many false anomalies chance produces.
  - then we correct the real tests with Bonferroni and Holm.

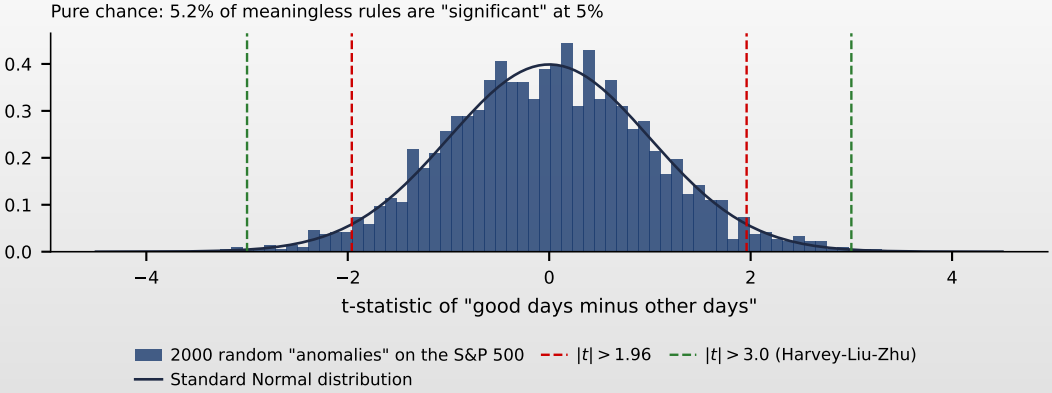
## Calendar effects and data snooping

- A calendar effect is a difference in mean returns associated with a day or a month.
- Day-of-the-week effect: mean returns differ across Monday, Tuesday, . . . , Friday French (1980).
- January effect: the mean January return differs from that of the other months Rozeff & Kinney (1976).
- Turn of the month (TOM): the last trading day of a month and the first three of the next Ariel (1987).
- Data snooping: selecting a pattern after searching many alternatives on the same data.

## How Many "Anomalies" Does Pure Chance Produce? (1/3)

- Why we simulate: here every null hypothesis is true, so every rejection is false.
- Experiment: 2 000 trading rules with no economic content, on daily S&P 500 returns.
- Each rule picks 20% of the days at random and calls them "good days".
- For each rule, a $t$ test compares the mean return on "good days" with the mean on the other days.
- The chart: the histogram of the 2 000 $t$ statistics, over the standard Normal density.
- The vertical lines: the thresholds $|t| = 1.96$ (5%) and $|t| = 3.0$, the bar proposed by Harvey, Liu & Zhu (2016).

# How Many "Anomalies" Does Pure Chance Produce? (2/3)




MFM_ch2_multiple_testing ▶ Interactive

## How Many "Anomalies" Does Pure Chance Produce? (3/3)

- Result: 5.2% of the rules are "significant" at 5% ($|t| > 1.96$), as the test level implies.
- The most "spectacular" rule has $|t| = 3.30$, although it is pure chance.
  - reported alone, it would pass even the strict bar $|t| > 3.0$.
- The bar $|t| > 3.0$ is passed by 0.30% of the random rules, i.e. 6 rules out of 2 000.
- How many rules were tried (the family of tests) matters as much as the bar chosen.
- Sullivan, Timmermann & White (1999) show how to correct for data snooping in trading rules.

MFM_ch2_multiple_testing

## Reading the calendar-effects table

**HAC**: Heteroskedasticity and Autocorrelation Consistent.

- We apply the same logic to 15 real tests, on daily data from 2000.
  - five markets $\times$ three effects (weekday, January, TOM).
- We compare developed markets (S&P 500, Euro Stoxx 50) with emerging markets of the region (BET, WIG20, BUX).
- January extra: the mean January return minus the mean of the other months, in percent per month.
- TOM extra: the mean return on TOM days minus the mean of the other days.
  - in basis points per day (bp; 1 bp $= 0.01$ percentage points).
- We use HAC errors because returns show volatility clustering.
  - HAC errors remain valid under it; ordinary ones do not.
  - In these tests the HAC errors are 0.92–1.54 times the ordinary ones: smaller as well as larger.

## January and Turn-of-the-Month Effects

| **Market** | January extra (%/month) | $t_{HAC}$ | TOM extra (bp/day) | $t_{HAC}$ |
|---|---|---|---|---|
| S&P 500 (US) | −0.66 | −0.80 | 3.2 | 0.92 |
| BET (Romania) | 3.48 | 1.49 | 17.3 | 3.62 |
| Euro Stoxx 50 | −0.30 | −0.29 | 5.1 | 1.21 |
| WIG20 (Poland) | 0.53 | 0.40 | 8.2 | 1.72 |
| BUX (Hungary) | 2.66 | 2.25 | 15.1 | 3.63 |

- TOM: the extra return is +17.3 bp per day for the BET and +15.1 bp for the BUX
  - both with $|t| > 3.5$, above the 3.0 bar of Harvey, Liu & Zhu (2016)
- For the S&P 500, the TOM extra return is only +3.2 bp per day and not significant ($t = 0.92$)
- For the BET, January exceeds the other months by 3.48 percentage points
  - but the estimate is imprecise ($t = 1.49$)
- The BUX, with $t = 2.25$, is the only market with a January-effect $p$-value below 0.05
- The third effect, the day of the week, is not significant in any market
  - The smallest $p$-value is 0.2265 (WIG20)
- We ran 15 tests (5 markets $\times$ 3 effects): which survive a correction for multiple testing?
- What the result supports: association

## Multiple Testing: Bonferroni and Holm

**FWER**: Family-Wise Error Rate — the probability of at least one false rejection in the family of tests.

- Why FWER: we want every anomaly we report to be real, not only most of them.
- With $M$ independent tests at level $\alpha$: FWER $= 1 - (1-\alpha)^M$.
- For $M = 15$ and $\alpha = 5\%$, FWER would be 53.7% if the tests were independent.
- **Bonferroni**: reject only if $p_i < \alpha/M$.
  - it guarantees FWER $\leq \alpha$ under any dependence between the tests.
- **Holm**: sort $p_{(1)} \leq \cdots \leq p_{(M)}$ and reject while $p_{(j)} < \alpha/(M-j+1)$, stopping at the first failure.
- Holm has the same guarantee and rejects at least everything Bonferroni rejects.
- Bootstrap methods that use the correlation between tests (StepM): Appendix.

## Check: multiple testing and anomalies

- Why does testing many rules change the meaning of a small p-value?
- Does non-rejection prove efficiency?
- Which new sample or robustness check would you request?

## Answers: multiple testing and anomalies

- Searching many rules creates more opportunities for false rejection.
- No. A test with low power may fail to reject even if the market is inefficient.
- The conclusion concerns only the alternative tested, not every form of predictability.
- Example: the same rule on post-publication data or on another market.
  - a data-snooping artefact disappears out of sample.
- A real effect usually weakens after publication but does not vanish McLean & Pontiff (2016).

## Main results of the chapter

- S&P 500: return reversal at horizons of 2–10 days (variance ratio below 1).
- BET: persistence (variance ratio above 1) at all four tested horizons.
- The dividend–price ratio does not beat the historical mean out of sample ($R^2_{OS} < 0$).
- Bank tax: the abnormal component of TLV's fall, measured against the BET, was almost offset over the next five sessions.
  - a single event cannot separate overreaction from later news.
- After the Holm correction only the TOM effects for the BET and the BUX remain significant.

## Topics for further study

- Data and conventions
- Fair games, pricing and rational bubbles
- Test construction and alternatives
- Predictive regressions and bias
- Long memory and changing efficiency
- Event-study inference
- Anomalies and behavioural explanations
- Machine learning and research design
- Case study: time-series momentum in Bitcoin
- AI for scientific discovery
- Synthesis and technical reference

# Data Used in This Chapter

**BNR**: Banca Națională a României — the National Bank of Romania.

- 16 daily series up to 18 September 2026
  - sample sizes: Appendix
  - 13 equity indices from 2000: 4 developed (S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng), 9 emerging or frontier (BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai)
  - Bitcoin from 2014 in the full-sample and rolling tests; the yearly crypto analysis starts in 2011
  - Ethereum from 2016 and EUR/RON, the BNR reference rate, from July 2005
- Daily log returns $r_t = \ln P_t - \ln P_{t-\mathbf{1}}$, each series on its own calendar
  - $P_t$: the closing price (index level, exchange rate) on day $t$
- Equity indices use trading days only
  - a repeated close is removed only if the day shows no trading (zero volume, or open = high = low = close)
  - or if its whole row copies the previous day's; such a day is a holiday filled with the previous price
  - a genuine unchanged close, with volume and an intraday range, stays as a zero return (S&P 500: 3 of about 6 700 days)
  - the BET file has closes only: none of its days is removed
  - IPC: 46 such days (44 in 2023–2026) with near-identical volume, likely provider duplicates, are kept
  - dropping them moves $\hat{\rho}_{\mathbf{1}}$ by 0.001, $z^*(2)$ by 0.04, $\hat{d}$ by 0.014, $AQ$ by 0.29
- Crypto-assets trade 7 days a week: 365 observations per year
- Analyses that combine several series align prices on common trading days
  - e.g. event studies
  - before computing returns

## What restriction does efficiency impose on prices?

- The question: what exact restriction does efficiency impose on prices, and what does it leave free?
- The standard answer is the fair game.
  - The return in excess of the required return cannot be forecast.
- The autocorrelation tests of section 2 check only one consequence of this condition.
- The required return comes from a pricing model, so every fair-game test is also a test of that model Fama (1991).
- A fair game allows rational bubbles.
  - but rules out deviations that revert predictably to fundamental value (fads).
- Deviations do not always disappear Shleifer & Vishny (1997).
  - arbitrage is risky when price convergence is uncertain (Further Topics).

## Moments, tails and asymptotics

- The question: which assumptions make an efficiency test valid, and which alternatives can it detect?
- Robust standard errors of autocorrelations (Further Topics) need a finite fourth moment, $\mathbb{E}[r^4] < \infty$
  - heavy tails: Student $t$ with four degrees of freedom has finite variance but $\mathbb{E}[r^4] = \infty$, and then the robust standard error no longer converges
- Critical values ($\pm 1.96$, $\chi^2$ quantiles) are asymptotic approximations
  - correct as the sample grows
  - possibly inaccurate in a short sample such as a 500-day window
- Limit theorems also need weak dependence (mixing)
  - observations far enough apart in time become nearly independent
  - mixing allows volatility clustering and dependence between neighbouring days

## Predetermined variables, bias and consistency

- Why does a persistent predictor distort inference in a finite sample?
- A predetermined predictor is known before the return shock it predicts
  - strict exogeneity additionally rules out correlation with shocks at every date.
- Today's dividend–price ratio can forecast next month's return.
  - yet next month's price shock also changes the ratio.
  - this feedback obstructs strict exogeneity.
- Consistent is not unbiased.
  - with a merely predetermined predictor, OLS converges to the true parameter.
  - but is biased in a finite sample.

## Short memory, long memory and scaling

- Does dependence persist across horizons and market regimes?

**AR**: AutoRegressive model.

- Short memory has absolutely summable autocovariances
  - an AR(1) with coefficient 0.5 has correlations $0.5^k$ at lag $k$.
- Long memory here means positive dependence decaying so slowly that the autocovariance sum diverges
  - power-law decay can produce this.
- Scaling: a measure of fluctuations over $n$ days, $F(n)$, grows like $n^H$
  - e.g. the standard deviation of a sum of $n$ returns
  - $H$ is the Hurst exponent, estimated in Further Topics
  - doubling the horizon multiplies fluctuations by $2^H$: about 1.41 at $H = 0.5$ (no long memory) and 1.68 at $H = 0.75$
- Anti-persistence describes offsetting increments, as in fractional Gaussian noise (fGn) with $H < 0.5$
  - fGn is a stationary Gaussian increment model with partial-sum variance proportional to $n^{2H}$.

## Variance Ratio and Hurst Exponent: Long or Short Memory?

- For fGn, the variance of a sum of $q$ returns is $\mathrm{Var}(r_t + \cdots + r_{t-q+1}) = q^{2H}\mathrm{Var}(r_t)$.
  - fGn: the exactly self-similar process defined above.
- Hence $\mathrm{VR}(q) = q^{2H-1}$, and an observed $\mathrm{VR}(q)$ implies a Hurst exponent $H(q) = 0.5 + \ln \mathrm{VR}(q)/(2 \ln q)$.
- At $q = 20$ days, the implied exponent is 0.553 for the BET and 0.453 for the S&P 500.
- The problem: a process without long memory can give $H(q) \neq 0.5$
  - Example: an AR(1) with $\rho_1 = -0.1$ has autocorrelations $\rho_k = (-0.1)^k$, so only short memory.
  - For this AR(1), $\mathrm{VR}(20) = 0.826$, so $H(20) = 0.468$: the process looks "anti-persistent" without any long memory.
  - At $q = 250$, the same process gives $H(250) = 0.482$: for short memory, $H(q)$ approaches 0.5 as $q$ grows.
- An AR(1) with BET's autocorrelation, $\hat{\rho}_1 = 0.109$, gives $H(20) = 0.535$.
- So most of the gap between BET's $H(20) = 0.553$ and 0.5 is short memory.
- Separating long from short memory needs a statistic built for it: Lo's modified R/S (Further Topics).

## Event-study inference

- The question: how reliable are the $t$ statistics of the Event studies section?
  - when their assumptions fail
- There we assumed:
  - $\hat{a}_i$ and $\hat{b}_i$ are known exactly.
  - firms are independent.
  - the abnormal-return variance does not rise on the event day.
- Estimation error: $\hat{a}_i$ and $\hat{b}_i$ come from a finite window.
  - the predicted normal return is also uncertain.
- Cross-sectional dependence: two banks hit by the same news have abnormal returns that move together.
- Event-induced variance: the event itself raises the variance.
  - the calm days before understate it.

## Anomalies and behavioural explanations

- Can behavioural biases explain the anomalies that survive?

**Momentum**

- Cross-sectional momentum ranks assets.
  - if A gained 10% and B gained 2%, a relative strategy may buy A and short B.
- Time-series momentum uses each asset's own past sign.
  - both A and B would receive positive signals in that example.

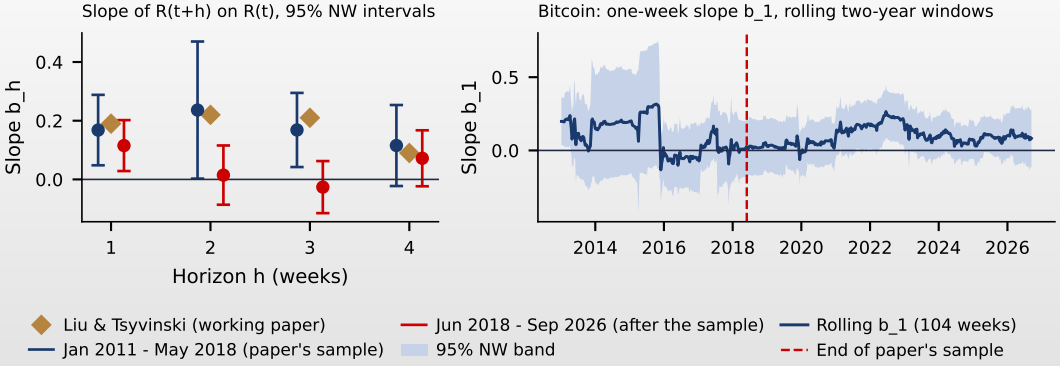
## Machine learning and research design

- The question: what does an in-sample rejection mean?
  - when we test efficiency with hundreds of predictors
- Anomalies are found by searching among many predictors: Harvey, Liu & Zhu (2016) count more than 300 published factors.
- The Multiple testing and anomalies section showed that 2 000 random rules produce in-sample "anomalies".
- Here:
  - how to evaluate a flexible model properly.
  - why an in-sample rejection can arise even with rational investors Martin & Nagel (2022).

## The Question: Do Past Returns Predict Bitcoin's Return?

- **Question**: does this week's Bitcoin return predict the returns of the following weeks?
  - if so, past prices contain unused information: a departure from weak-form efficiency
- Liu & Tsyvinski (2021), *Review of Financial Studies*
  - cryptocurrency returns have no exposure to the usual stock market factors or to macroeconomic factors
  - they are predicted by factors specific to crypto markets: time-series momentum (TSMOM) and investor attention
- TSMOM: a week with a high return is followed, on average, by high returns in the following weeks
- The authors' numbers (working paper of the National Bureau of Economic Research, NBER w24877; Bitcoin, January 2011 – May 2018)
  - slope of the regression of next week's return on this week's return: 0.19 ($t = 3.73$)
  - one more standard deviation today (16.6%) adds 3.2% to next week's return
  - the quintile of weeks with the highest returns is followed by 11.2% a week on average, the one with the lowest by 2.6%
- Why it matters: an effect that investors exploit should weaken after publication McLean & Pontiff (2016)

# Bitcoin: The Momentum Slope in the Paper's Sample and After




- Left: the slope $b_h$ in $R_{t+h} = a + b_h R_t + e_{t+h}$, with 95% Newey–West (NW) intervals; diamonds: the authors' numbers
- Right: the slope $b_1$ over rolling windows of 104 weeks (two years), dated at the last week of the window

MFM_ch2_crypto_momentum ▶ Interactive

## The Method in Brief and How the Replication Differs

- $R_t$: the simple return of Bitcoin in week $t$, from the Sunday close to the next Sunday close
  - daily closing prices in UTC (Coordinated Universal Time), as in the paper
- Predictive regression $R_{t+h} = a + b_h R_t + e_{t+h}$, for $h = 1, \ldots, 4$ weeks
  - the paper's sample: 9 Jan 2011 – 27 May 2018, 386 weeks
  - after the sample: 3 Jun 2018 – 13 Sep 2026, 433 weeks
  - the authors' robustness check: the same regression from 2013
- Sorting: the weeks are split into five quintiles by $R_t$; we compare the mean of $R_{t+1}$ in the top and the bottom quintile
- Differences from the paper
  - EODHD (EOD Historical Data) prices from the course data instead of CoinDesk prices
  - NW standard errors with 5 lags instead of the authors' OLS (ordinary least squares) and bootstrap standard errors
  - no attention factors (Google searches, Twitter) and no Ripple

# The Result: Only the One-Week Effect Remains, Three Times Smaller

- In the paper's sample, the course data reproduce the result
  - $b_1 = 0.168$ (OLS $t = 3.33$, NW $t = 2.75$), against 0.19 in the paper
  - the top quintile is followed by 9.9% a week, the bottom one by 2.6% ($t$ of the difference $= 2.18$)
- After the sample, predictability remains only at the one-week horizon
  - $b_1 = 0.115$ (NW $t = 2.61$); it survives the Holm correction over the four horizons ($p = 0.009 < 0.05/4$)
  - at $h = 2, 3, 4$: slopes between $-0.026$ and $0.072$, all with $|t| \leq 1.48$
  - one more standard deviation today adds 0.96% to next week's return, against 2.98% in the paper's sample
  - weekly volatility fell from 17.7% to 8.3%; Ethereum after the sample: $b_1 = 0.107$ (NW $t = 2.45$)
- What the result supports: statistical predictability of the one-week return, not a profitable strategy
  - no trading costs, no risk adjustment and no out-of-sample forecasts
- Limits
  - from 2013, within the paper's sample, $b_1$ is no longer significant (NW $t = 1.06$): the extreme weeks of 2011–2012 weigh heavily
  - $b_1$ is significant in only 13% of the rolling two-year windows
  - the NW $t$-test requires a finite fourth moment, which Chapter 1 questions

## An Open Question: Does a Market Become More Efficient When Arbitrageurs Enter?

- The question: does Bitcoin's informational efficiency rise when new arbitrageurs enter?
  - the new arbitrageurs come in through regulated instruments
  - such as spot ETFs
- Tran & Leirvik (2020) find that the efficiency of crypto-assets varies over time and rises.
- Kia, Liu, Li, Song & Xu (2026) find that, after the spot ETFs, the spot market again became the dominant venue of price discovery.
- We do not know whether the change comes from the ETFs or from market maturity.
  - this is the question tested in this section.


▸ Appendix: ETFs, placebo

# Mini-Case: Did Spot ETFs Make Bitcoin More Efficient?

- An artificial-intelligence (AI) assistant proposes a typical hypothesis:
  - "US spot Bitcoin ETFs (trading since 11 January 2024) brought arbitrageurs, so Bitcoin became more efficient"
- The discovery loop
  - reusable prompts: Appendix
  - **literature**: Kia, Liu, Li, Song & Xu (2026) and Tran & Leirvik (2020) (the topic's opening slide)
  - **hypothesis**: $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$, two windows of equal length around the launch date (first trading day)
  - **data and code**: daily Bitcoin closing price (USD); first replicate the chapter's full-sample $\hat{\rho}_1 = -0.022$
- Checks fixed before seeing results
  - rationale for each: Appendix
  - HAC standard errors of the estimates (not $1/\sqrt{T}$); equal windows; the launch date only, plus placebo dates
  - a control asset for confounders (the April 2024 halving, rates, inflows)
  - a multiple-testing correction (Holm or Chow–Denning) for every statistic tried, and the minimum detectable effect (MDE)


Appendix: mini-case results

## Your Turn: A Project Seed

- **Question**: did the spot ETFs change the time-varying efficiency of Bitcoin?
  - relative to a control asset
  - rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
  - if parallel trends, no anticipation and no spillovers are not credible, report descriptive, not causal, differences
- **Replicate first**: the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
  - record which Bitcoin price series and first date each window uses, placebo windows included
- **Pre-register**: $H_0$, $H_1$, dates, windows and corrections, committed before the results
- **Report**: robust inference, placebo tests and the MDE
- Publish the code as a Quantlet
- **Log**: record the prompts used in `AI_USE.md`
- Record in `AI_ERRORS.md` every error of the AI assistant you caught
  - for example an i.i.d. standard error or an invented reference

# Synthesis, Check Yourself, and Next: Chapter 3

## Five steps for every efficiency claim

- **Information set**: which information $\mathcal{I}_t$ do we assume prices reflect (past prices, public news)?
- **Null**: RW1, RW3 or the martingale difference, stated together with the risk model
- **Statistic**: its variance valid under volatility clustering
- **Uncertainty**: horizons, markets and windows tried, and out-of-sample evidence
- **Economics**: a statistical rejection is not an exploitable profit after costs
  - a profit does not prove an irrational market

## Check yourself and next steps

- Why can RW3 be rejected for the S&P 500 while the market remains efficient in the economic sense?
- Why does BET's $H(20) > 0.5$ not prove long memory?
- Which of the two results would survive a change of test, window and inference?
  - BET's persistence or Bitcoin's non-rejection
- Next, Chapter 3: risk-adjusted returns with factor models
  - the factor zoo
- Textbooks: Campbell, Lo & MacKinlay (1997), ch. 2, and Franke, Härdle & Hafner (2019)
  - advanced courses: ▸ Appendix

## The Practical Decision

- **What I choose**
  - weak-form tests with robust standard errors, not i.i.d. ones (S&P 500: robust SE of $\hat{\rho}_1$ 0.027, against 0.012)
  - Chow–Denning across the four variance-ratio horizons, Holm across the 15 calendar tests
  - for a predictor: $R^2_{OS}$ and the Clark–West test against the historical mean, not the in-sample $R^2$
- **Under which conditions**
  - the robust standard errors require a finite fourth moment, which Chapter 1 questions
  - a rejection also tests the model of expected returns and is not a profit after costs
- **What result would make me drop it**
  - a predictor: $R^2_{OS} < 0$ with an insignificant Clark–West test, as *dp* on the S&P 500 ($-0.51\%$, $t = 0.42$)
  - an anomaly: a rejection that does not survive Holm, as the BUX January effect ($t = 2.25$)
  - the BET persistence: a variance ratio that stays inside the robust band on new windows

## Why the robust band is wider: a numerical check

- For $q = 2$, the classical statistic is $z = \sqrt{T}\,(\widehat{\mathrm{VR}}(2) - 1)$.
  - it assumes i.i.d. returns (RW1).
- The robust statistic $z^*$ divides the same numerator by $\sqrt{\hat{\theta}(2)}$.
- Without volatility clustering, $\hat{\theta}(2) \approx 1$ and $z^* \approx z$.
  - with volatility clustering, $\hat{\theta}(2) > 1$ and $|z^*| < |z|$.
- Toy example: $T = 400$ days and $\widehat{\mathrm{VR}}(2) = 0.90$ give $z = 20 \times (-0.10) = -2.0$: the null is rejected at 5%.
- If volatility clustering makes $\hat{\theta}(2) = 4$, then $z^* = -2.0/\sqrt{4} = -1.0$ and the null is no longer rejected.
- The robust band around 1 is then $\sqrt{4} = 2$ times wider.
- For the S&P 500, $z = -8.15$ and $z^* = -3.66$.
  - the robust statistic is about 2.2 times smaller.
  - the null is still rejected.

## Why the dividend–price ratio?

- The Campbell & Shiller (1988) identity: $dp_t \approx c + \sum_{j \geq 0} \rho^j \left( r_{t+1+j} - \Delta d_{t+1+j} \right)$.
  - $\rho$ just below 1.
  - $\Delta d$: log dividend growth.
- The identity follows from the definition of the return through a log-linear approximation.
  - with no economic model.
- Consequence: if $dp$ varies, it must forecast future returns, future dividend growth or both.
- In the data $dp$ forecasts dividend growth poorly, so its variation should forecast returns Cochrane (2008).
- That is why $dp$ is the most studied market predictor Fama & French (1988).
- A successful forecast would not prove inefficiency.
  - it can reflect a risk premium that varies over time.

## Holm on the 15 calendar p-values

- Two-sided $p$-values:
  - the HAC $t$ statistic against the Normal distribution (January, TOM).
  - the Wald $\chi^2_4$ test that the five daily means are equal (weekdays).

| **Market** | Weekday, equal means | January | TOM |
|---|---|---|---|
| S&P 500 (US) | 0.8263 | 0.4257 | 0.3585 |
| BET (Romania) | 0.3052 | 0.1368 | 0.0003 |
| Euro Stoxx 50 | 0.6319 | 0.7738 | 0.2266 |
| WIG20 (Poland) | 0.2265 | 0.6865 | 0.0853 |
| BUX (Hungary) | 0.9327 | 0.0245 | 0.0003 |

- The two smallest $p$-values are TOM for the BET and TOM for the BUX, both 0.0003.
- They are below the Holm thresholds $0.05/15 = 0.0033$ and $0.05/14 = 0.0036$, so both nulls are rejected.
- The third $p$-value, January for the BUX (0.0245), exceeds the threshold $0.05/13 = 0.0038$, so Holm stops here.
- Without correction, January for the BUX would have been the third rejection, since $0.0245 < 0.05$.

## Market abbreviations

- **US**: United States
- **HK**: Hong Kong
- **BIST**: Borsa İstanbul — the Istanbul Stock Exchange (BIST 100 index).
- **WIG20**: Warszawski Indeks Giełdowy 20 — the Warsaw Stock Exchange index of the 20 largest companies.
- **BUX**: Budapesti Értéktőzsde index — the Budapest Stock Exchange index.
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha).
- **IPC**: Índice de Precios y Cotizaciones — the Mexican Stock Exchange index.
- **BET**: Bucharest Exchange Trading (index).
- **FX**: Foreign eXchange.

## Fair games and martingales

- **Fair game** for excess returns: $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$
  - $\mathcal{I}_t$: the information available at the end of day $t$; $\mathbb{E}[\cdot \mid \mathcal{I}_t]$: the expectation given that information
  - $\mu_t$: the required (equilibrium) return, the compensation for risk
  - in words: the return beyond $\mu_t$ cannot be forecast from $\mathcal{I}_t$
- **Martingale**: a process whose expected next value, given the past, equals its current value
  - with a constant required simple return $m$, the value $V_t$ of a position with dividends reinvested, discounted at $m$, $V_t/(1+m)^t$, is a martingale Campbell, Lo & MacKinlay (1997)
  - discounting at the risk-free rate gives a martingale only under the risk-neutral measure (slide on the risk-neutral measure)
  - the price $P_t$ itself is a martingale only if no dividends are paid and $m = 0$; the expected log return is then negative (the Jensen term, next slide)

## The Jensen term: why averages differ

- The logarithm is concave.
  - averaging log outcomes gives less than taking the log of their average, unless outcomes are constant.
- Equally likely simple returns of $+10\%$ and $-10\%$ have mean zero.
- The mean log return is $\frac{1}{2}\ln(1.1) + \frac{1}{2}\ln(0.9) = \frac{1}{2}\ln(0.99) \approx -0.00503$.
- Along a path with one rise and one fall, wealth 100 becomes $100(1.1)(0.9) = 99$.
  - a zero arithmetic mean does not preserve compounded wealth.

## Prices, log prices and information

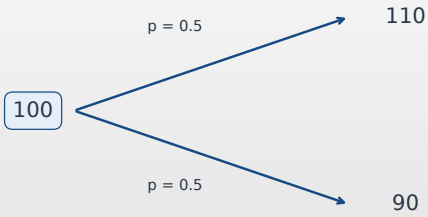
- A martingale in prices and a constant mean log return are different restrictions
  - a martingale in prices: $\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$
  - a constant mean log return: $\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$
  - the difference is the Jensen term: $\mathbb{E}[\ln(1+R)] \approx \mathbb{E}[R] - \mathrm{Var}(R)/2$, with $R = P_{t+1}/P_t - 1$ the simple return
- Consequence of the fair game: $\mathrm{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$ for any function $g$ of past information
  - where $\varepsilon_{t+1} = r_{t+1} - \mu_t$ is the unexpected part of the return
- If $\mu_t = \mu$ is constant, all autocorrelations $\rho_k = \mathrm{corr}(r_t, r_{t-k})$, $k \geq 1$, are zero
  - this is the hypothesis tested in section 2
- The converse fails: zero autocorrelations do not rule out a nonlinearly predictable mean
  - the MDH, next topic

## Discounting and the risk-neutral measure

- Why: the martingale above uses the actual probabilities and the required return $m$.
  - pricing theory uses, equivalently, the risk-free rate and risk-adjusted probabilities.
- A risk-neutral measure uses pricing probabilities, in an arbitrage-free model.
  - under which discounted traded prices, including dividends, are martingales.
- Example: a stock without dividends costs 100 today.
  - in one year it is worth 120 or 80.
  - the risk-free rate is 5%.
  - The pricing probability $q = 0.625$ gives $q\,120 + (1-q)80 = 105$, and $105/1.05 = 100$, today's price.
- This $q$ need not equal the actual probability of an increase.
  - pricing probabilities are not forecasts of frequencies.

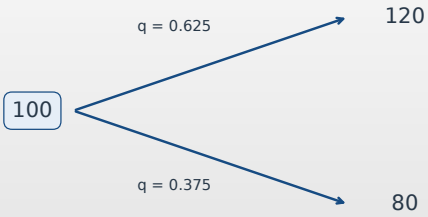
# Probabilities and conditional means

**Martingale under actual probabilities**




Next-value mean = 100
Expected increment = 0

**Risk-neutral pricing example**




Next-value mean = 105
Discounted: 105 / 1.05 = 100

Left: the actual conditional mean stays at 100. Right: pricing probabilities give 105, whose discounted value is 100.

## Fundamental value, rational bubbles and fads

- With a constant required simple return $m$, the fair game is $\mathbb{E}[P_{t+1} + D_{t+1} \mid \mathcal{I}_t] = (1+m)\,P_t$
  - $D_{t+1}$: the cash flow (dividend) paid in $t+1$
  - Fundamental value $P_t^* = \mathbb{E}\big[\sum_{j\geq 1} D_{t+j}/(1+m)^j \mid \mathcal{I}_t\big]$; pricing error $B_t = P_t - P_t^*$
  - The fundamental value $P_t^*$ satisfies the same fair-game equation as the price
  - Subtracting the two equations, the gap $B_t$ satisfies $\mathbb{E}[B_{t+1} \mid \mathcal{I}_t] = (1+m)\,B_t$
- A fair game ties the price to the fundamental value only up to a **rational bubble**
  - A gap $B_t > 0$ sustained by the expectation of resale
  - A gap $B_t \neq 0$ is admissible only if it is expected to grow at the required rate $m$ (Chapter 17)
  - A mean-reverting gap (fad) makes $\varepsilon_{t+1}$ predictable and violates the fair game, although short-horizon tests have little power against it Summers (1986)
- The restriction is on the conditional mean only
  - The variance is not restricted
  - volatility may be predictable (Chapter 1)

## Rational bubble: a numerical example

- A share pays 10 RON at the end of every year, forever.
  - The first dividend is paid one year from today.
  - The constant required return is $m = 10\%$; fundamental value is $P_t^* = 10/0.10 = 100$ RON.
- Today $P_t = 120$ RON, so $B_t = 20$ RON.
  - Without a crash, $B_{t+1} = 1.1 \times 20 = 22$ RON.
  - After next year's dividend, fundamental value remains 100 RON; the price is therefore 122 RON.
- The total simple return is $(122 + 10 - 120)/120 = 10\%$.

## Rational bubble: what happens if it can burst?

- Initial bubble $B_t = 20$ RON and required return $m = 10\%$.
  - Survival probability $\pi = 0.8$; crash probability $1 - \pi = 0.2$.
  - A crash makes the bubble zero; investors know the probabilities.
- The fair game for the bubble, $\mathbb{E}[B_{t+1} \mid \mathcal{I}_t] = (1+m)B_t$, requires an expected bubble of $(1+m)B_t = 22$ RON.
  - Conditional on survival: $B_{t+1} = 22/0.8 = 27.5$ RON.
  - Conditional growth is $27.5/20 - 1 = 37.5\%$.
- Crash mechanism: Blanchard & Watson (1982), Bubbles, Rational Expectations and Financial Markets.

- The purchase price is 120 RON.
  - the dividend at year-end is 10 RON.
- Survival: the ex-dividend price is 127.5 RON.
  - Simple return: $(127.5 + 10 - 120)/120 \approx 14.6\%$.
- Crash: the price falls to fundamental value, 100 RON.
  - Simple return: $(100 + 10 - 120)/120 \approx -8.3\%$.
- Expected return: $0.8 \times (127.5 + 10 - 120)/120 + 0.2 \times (100 + 10 - 120)/120 = 10\%$.
  - The expected return satisfies the fair-game condition; either realised return can differ from it.

## Rational bubble and the no-bubble condition

- The fair game restricts only the expected return over the next period.
  - it does not pin down the price level.
- **No-bubble condition**: $\lim_{k\to\infty} \dfrac{\mathbb{E}[P_{t+k} \mid \mathcal{I}_t]}{(1+m)^k} = 0$
  - also called the transversality condition
  - $k$: the resale horizon; the discounted expected resale price vanishes
- Under this condition the price equals the fundamental value $P_t^*$.
  - it is supported entirely by dividends.
- A rational bubble violates the condition.
  - from $\mathbb{E}[B_{t+k} \mid \mathcal{I}_t] = (1+m)^k B_t$, the discounted value of the bubble stays $B_t \neq 0$ at every horizon.
- In the numerical example, the 20 RON bubble is still worth 20 RON in today's money, however distant the resale.
- Tirole (1985): in overlapping-generations models, a bubble can exist only if:
  - the interest rate is below the growth rate of the economy.

## The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it Grossman & Stiglitz (1980)
  - Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**
  - just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency
  - a prediction for our comparison of developed, emerging and crypto markets
- Efficiency does not rule out bubbles either
  - it says they are hard to time (Chapter 17)

## Pure arbitrage and relative-value trading

- Pure arbitrage is a strategy with:
  - no positive initial outlay.
  - no possible loss.
  - a positive gain in at least one state.
- Relative-value trading is risky when price convergence is uncertain.
- Example: a price gap of 20 points may widen to 30 before it closes.
  - the position then needs extra collateral (margin) and may be liquidated.
- The link with efficiency: arbitrageurs with limited capital cannot eliminate deviations from fundamental value Shleifer & Vishny (1997).
  - when convergence is uncertain.

## Autocorrelation and Its Robust Standard Error

- Sample autocorrelation: $\hat{\rho}_k = \sum_{t=k+1}^{T} e_t e_{t-k} / \sum_{t=1}^{T} e_t^2$, with $e_t = r_t - \bar{r}$
  - $T$: number of daily returns; $\bar{r}$: their mean; $k$: the lag; $\hat{\rho}_k > 0$: persistence, $\hat{\rho}_k < 0$: reversal
- Under RW1 (i.i.d.): $\mathrm{SE}(\hat{\rho}_k) \approx 1/\sqrt{T}$; 95% band $\pm 1.96/\sqrt{T}$
- Unpredictable mean with volatility clustering (finite fourth moments): $\mathrm{SE}(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$
  - $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / \left(\sum_t e_t^2\right)^2$ (robust variance under $H_0$; for test bands centred at 0)
  - When large moves cluster, $\hat{\tau}_k > 1/T$: the i.i.d. band is too narrow and classical tests reject too often; the robust statistics $\tilde{Q}$ (Ljung–Box) and $z^*$ (variance ratio) use $\hat{\tau}_k$ instead of $1/T$

## Intervals for $\rho_k$: the HAC Standard Error

- $\sqrt{\hat{\tau}_k}$ is derived under $H_0$: valid for a test band centred at 0, not for an interval around $\hat{\rho}_k \neq 0$.
- Around the estimate we use the HAC variance of the estimator Newey & West (1987):
  - $\hat{\psi}_t = (e_t e_{t-k} - \hat{\rho}_k e_t^2)/\hat{\gamma}_0$, with $\hat{\gamma}_0 = \frac{1}{T}\sum_t e_t^2$: the contribution of day $t$ to $\hat{\rho}_k - \rho_k$.
  - $\widehat{\mathrm{Var}}(\hat{\rho}_k) = \frac{1}{T}\left[\hat{c}_0 + 2\sum_{j=1}^{\ell}\left(1 - \frac{j}{\ell+1}\right)\hat{c}_j\right]$, with $\hat{c}_j$ the lag-$j$ autocovariance of $\hat{\psi}_t$.
  - bandwidth $\ell = \lfloor 4(T/100)^{2/9} \rfloor$: 9–10 lags for the full samples, 4–5 for one calendar year, 6 for the ETF windows.
- Use: intervals around $\hat{\rho}_k$ and comparisons between periods (crypto years, before/after the ETFs).
- Check: a moving-block bootstrap (blocks of $T^{1/3}$ days) gives similar values.
  - S&P 500: HAC 0.030, block bootstrap 0.031, $\mathrm{SE}_0 = 0.027$.
- The HAC SE is smaller than $\mathrm{SE}_0$ for 14 of the 16 series (BET: 0.023 against 0.028).

## Joint tests and reference distributions

- A portmanteau test combines several lag correlations into one statistic.
  - instead of selecting the most striking lag.
- A Wald test similarly tests coefficient restrictions jointly
  - five equal weekday means impose four restrictions, $\beta_1 = \beta_2$, $\beta_1 = \beta_3$, $\beta_1 = \beta_4$, $\beta_1 = \beta_5$, so the reference is $\chi^2_4$, with 5% critical value 9.49.
- Five restrictions, all $\beta_d = 0$, would test a different null: every weekday mean equal to zero.

## Portmanteau Tests: Ljung–Box and Its Robust Version

- Ljung & Box (1978): $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k) \sim \chi_m^2$ under i.i.d. returns
  - Tests jointly $H_0$: $\rho_1 = \cdots = \rho_m = 0$
  - $m$: the number of lags
- Robust version Lobato, Nankervis & Savin (2001): $\tilde{Q}(m) = \sum_{k=1}^{m} \hat{\rho}_k^2/\hat{\tau}_k \to \chi_m^2$ under RW3
  - if the $\hat{\rho}_k$ at different lags are asymptotically uncorrelated
  - the condition, $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j}\varepsilon_{t-k}] = 0$ for $j \neq k$, holds for a GARCH with symmetric innovations
- Three markets at $m = 10$ (5% critical value 18.31)
  - S&P 500: $Q = 97.1$ but $\tilde{Q} = 20.0$ ($p = 0.029$)
  - BET: $Q = 108.0$, $\tilde{Q} = 24.4$ ($p = 0.007$)
  - Bitcoin: $Q = 20.2$ ($p = 0.028$) but $\tilde{Q} = 11.9$ ($p = 0.294$)
- The robust test keeps the rejections for S&P 500 and BET, but not for Bitcoin
  - $\tilde{Q}$ requires a finite fourth moment, which Chapter 1 questions

## The Runs Test

- The runs test Wald & Wolfowitz (1940) uses only the **signs** of returns (zero-return days dropped), so heavy tails do not affect it
- A **run** is a maximal block of equal signs: ++|-|+++ has 3 runs
- With $n_1$ positive days, $n_2$ negative days and $n = n_1 + n_2$, if the signs are independent:
  - $\mathbb{E}[R] = \dfrac{2n_1n_2}{n} + 1, \quad \mathrm{Var}[R] = \dfrac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}, \quad z = \dfrac{R - \mathbb{E}[R]}{\sqrt{\mathrm{Var}[R]}} \approx N(0,1)$
  - $R$: the observed number of runs; $|z| > 1.96$ rejects independence at 5%
  - the test assumes a constant probability of a positive day
- Too **few** runs ($z < 0$) mean signs that persist
  - too **many** ($z > 0$) mean signs that alternate
- S&P 500: $z = 4.34$, signs alternate, as $\hat{\rho}_1 < 0$
- BET: $z = -6.05$, signs persist, as $\hat{\rho}_1 > 0$
- Bitcoin: $z = 2.68$ rejects independent signs although $\hat{\rho}_1$ is not significant
  - signs and magnitudes carry different information
  - all markets: Appendix

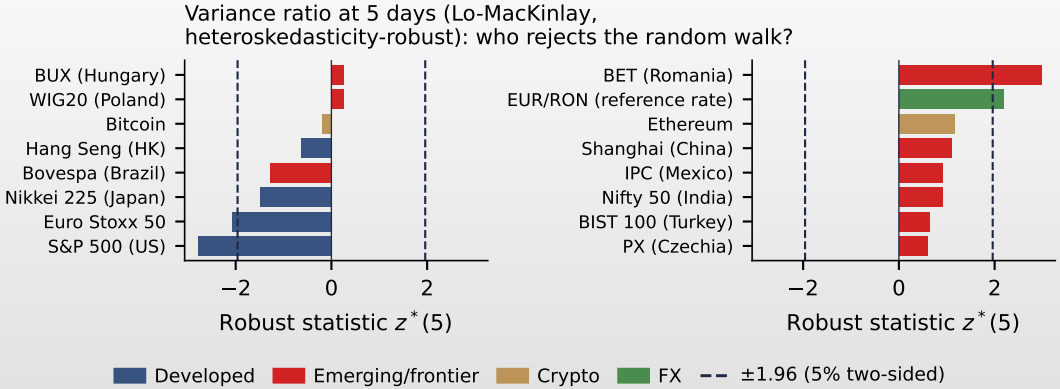
## The Lo–MacKinlay Test

- Lo & MacKinlay (1988) estimate $\mathrm{VR}(q)$ from overlapping sums of $q$ daily returns and correct the small-sample bias
  - $\hat{\sigma}_a^2 = \frac{1}{T-1}\sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m}\sum_{t=q}^{T}\left(\sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu}\right)^2$
  - $m = q(T-q+1)(1-q/T); \quad \widehat{\mathrm{VR}}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$
  - $\hat{\mu}$: mean daily return; $\hat{\sigma}_a^2$: variance of one-day returns; $\hat{\sigma}_c^2(q)$: variance of the overlapping $q$-day sums divided by $q$; $m$: a divisor that removes the small-sample bias
- Test statistic under RW1 (homoskedastic returns, i.e. with constant variance)
  - $z(q) = \dfrac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0,1)$
  - the denominator is the standard error of $\widehat{\mathrm{VR}}(q)$ under RW1; $|z| > 1.96$ rejects $\mathrm{VR}(q) = 1$ at 5%
- This $z$ rejects too often under volatility clustering (next slide)

## The Heteroskedasticity-Robust Statistic $z^*$

- Under RW3 plus the conditions of Lo & MacKinlay (1988), the $\hat{\rho}_k$ are asymptotically uncorrelated with variances $\hat{\delta}_k / T$
  - finite moments, weak dependence (mixing), $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$ for $j \neq k$
  - $\hat{\delta}_k = T\hat{\tau}_k$, with $\hat{\tau}_k$ of the autocorrelation slide: the robust variance of $\sqrt{T}\hat{\rho}_k$, equal to 1 without volatility clustering
- Asymptotic variance of $\widehat{\mathrm{VR}}(q)$ and the test statistic
  - $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left( \frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \qquad z^*(q) = \frac{\sqrt{T}\left(\widehat{\mathrm{VR}}(q) - 1\right)}{\sqrt{\hat{\theta}(q)}} \approx N(0,1)$
  - $2(q-k)/q$: the weight of $\hat{\rho}_k$ in $\widehat{\mathrm{VR}}(q)$; $|z^*| > 1.96$ rejects $\mathrm{VR}(q) = 1$ at 5%
- RW3 alone is not enough
  - the conditions hold for GARCH with symmetric innovations
  - but can fail with a leverage effect
  - Without the condition $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$, $\hat{\theta}(q)$ needs the full covariance matrix of the $\hat{\rho}_k$
- At $q = 5$, the S&P 500 has $z^*(5) = -2.78$
  - the reversal stays significant with the robust standard error
  - the comparison of $z$ with $z^*$ at $q = 2$: section 2

# The Variance Ratio at 5 Days on 16 Series




MFM_ch2_variance_ratio

## The 5-Day Variance Ratio and the Chow–Denning Test

- Each bar on the previous chart is the robust statistic $z^*(5)$ of one of the 16 series.
  - the dashed lines mark $\pm 1.96$.
- At 5 days, $|z^*(5)| > 1.96$ for:
  - the BET (2.98), the S&P 500 ($-2.78$).
  - EUR/RON (2.19) and the Euro Stoxx 50 ($-2.08$).
- The joint Chow & Denning (1993) test rejects RW3 for 5 of the 16 series.
  - critical value 2.49 (section 2).
  - BET, S&P 500, EUR/RON.
  - IPC (at 2 days, $z^* = 3.72$) and Ethereum (at 20 days, $z^* = 2.70$).
- The Euro Stoxx 50 rejects at 5 days but not in the joint test: its largest $|z^*|$ is $2.24 < 2.49$.
- No ranking "developed = efficient": the S&P 500 shows reversal, the BET and EUR/RON persistence.

MFM_ch2_variance_ratio

## Beyond Linear Predictability: the Martingale Difference Hypothesis

- **MDH** (Martingale Difference Hypothesis): $\mathbb{E}[r_t \mid \mathcal{I}_{t-1}] = \mu$
  - no function of the past predicts the mean, linear or not
- RW3 (all $\rho_k = 0$) is weaker. Counterexample: $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$, $\varepsilon_t$ i.i.d. symmetric
  - $\sigma^2 = \mathrm{Var}(\varepsilon_t)$, so the extra term has mean 0; $\theta \neq 0$: how strongly yesterday's squared shock moves today's mean
  - all $\rho_k = 0$, yet $\mathbb{E}[r_t \mid \varepsilon_{t-1}] = \theta(\varepsilon_{t-1}^2 - \sigma^2)$ is predictable: autocorrelation tests cannot see it
  - The MDH tests on the next slide look for exactly this kind of dependence (results: Appendix)
- **BDS** (Brock–Dechert–Scheinkman) Brock, Dechert, Scheinkman & LeBaron (1996) compares the frequency of nearby return sequences
  - with that implied by i.i.d. observations
  - it tests i.i.d., not the MDH
  - GARCH alone makes it reject
  - A BDS rejection is evidence against RW1, not against weak-form efficiency

## What nonlinear tests examine

- Why: the counterexample on the previous slide shows that zero autocorrelations do not rule out a predictable mean
- The generalised spectral test Escanciano & Velasco (2006) measures the covariance between $r_t$ and nonlinear functions of past returns
  - $e^{\mathrm{i}xr_{t-j}}$, at all lags $j$
  - it detects the counterexample $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$, which autocorrelations miss
- The automatic portmanteau test Escanciano & Lobato (2009) is a heteroskedasticity-robust Ljung–Box
  - with the number of lags chosen from the data by a penalty
  - it tests autocorrelations only, so it does not detect the nonlinear counterexample

## HAC standard errors and bandwidth

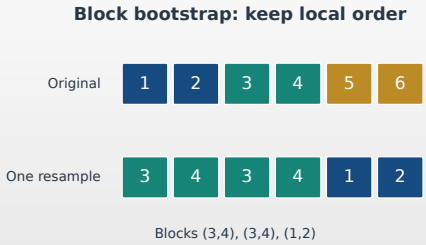
**OLS**: Ordinary Least Squares.

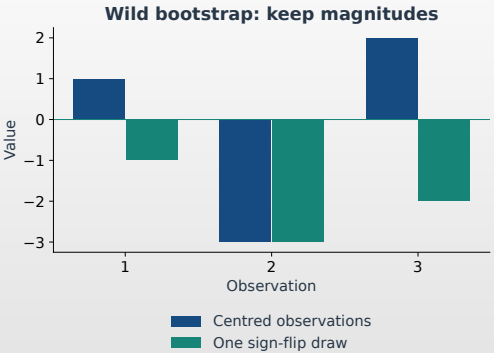
- Why: the residuals are autocorrelated and heteroskedastic
  - in predictive regressions and on overlapping rolling windows
  - and classical OLS standard errors are then not valid
  - in the dividend–price regression they are too small (OLS $t = 1.16$, Newey–West $t = 0.91$)
  - HAC errors can also be smaller than OLS ones (the calendar-effects table)
- HAC standard errors estimate the long-run variance
  - the variance plus twice the sum of the autocovariances
- Newey & West (1987) weight the autocovariance at lag $j$ by $1 - j/(\ell + 1)$, so that the estimate is never negative
- The bandwidth $\ell$ is the number of lags included
  - with $\ell = 5$ the weights of lags 1–5 are $5/6, 4/6, \ldots, 1/6$

## Bootstrap: constructing repeated samples

- Why: neither the Stambaugh bias nor the DFA null band has a known exact distribution
  - the bootstrap approximates it
- The bootstrap generates many artificial samples under the null and recomputes the statistic in each
- The i.i.d. bootstrap draws observations with replacement
  - and destroys all dependence, volatility clustering included
- The block bootstrap Künsch (1989) draws consecutive blocks and keeps the dependence within each block
- The wild bootstrap multiplies each demeaned return by a random sign
  - it keeps each day's volatility but destroys serial correlation: it reproduces an RW3-type null with symmetric signs (used for DFA)

# What each bootstrap preserves




Blocks preserve local ordering; sign flips preserve absolute magnitudes. They impose different assumptions on the artificial samples.

## Predictive Regressions and the Stambaugh Bias (1/2)

- Back to the regression of section 3: why can *dp* look like a good predictor even when $\beta = 0$?
- Model: $r_{t+1} = \alpha + \beta x_t + u_{t+1}, \quad x_{t+1} = \theta + \rho x_t + v_{t+1}$, with $\mathrm{Cov}(u, v) = \sigma_{uv}$
  - $r$: excess return; $x$: log dividend–price ratio $dp_t = \ln D_t - \ln P_t$, persistent ($\rho \approx 1$)
  - $D_t$: dividends paid over the last year; $\beta$: the predictive slope, 0 if returns are unpredictable; $\alpha$, $\theta$: intercepts
  - $\rho$: the persistence of $x$, the AR(1) coefficient of *dp* (not the return autocorrelation $\rho_k$, nor the $\rho$ of the Campbell–Shiller identity); $u$, $v$: the shocks of the two equations; $\sigma_{uv}$: their covariance; $\sigma_v^2 = \mathrm{Var}(v)$; $T$: number of periods

# Predictive Regressions and the Stambaugh Bias (2/2)

- $x_t$ is predetermined, not strictly exogenous: OLS is consistent but biased Stambaugh (1999)
  - Project $u$ on $v$: $u = (\sigma_{uv}/\sigma_v^2)\, v + e$, so $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$; $e$: the part of $u$ unrelated to $v$
  - Both regressions use the same months and the same $x_t$, so the estimation errors $\hat{\beta} - \beta$ and $\hat{\rho} - \rho$ are linked through the correlation of the shocks $u$ and $v$; the noise term has mean zero when $e$ is independent of the whole path of $x$, e.g. $(u, v)$ i.i.d. with a bivariate Normal distribution
  - $\hat{\rho}$ is biased downward in small samples Kendall (1954), and this error passes into $\hat{\beta}$
  - With $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$: $\quad \mathbb{E}[\hat{\beta} - \beta] \approx -\dfrac{\sigma_{uv}}{\sigma_v^2}\,\dfrac{1 + 3\rho}{T}$
- Price up $\Rightarrow$ $dp$ down: $\sigma_{uv} < 0$, so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the $t$-statistic is not $N(0, 1)$ either
  - valid inference by a Bonferroni $Q$-test over a confidence interval for $\rho$ Campbell & Yogo (2006)

## Bonferroni Q: allowing for uncertain persistence

- Campbell–Yogo inference allows for uncertainty about predictor persistence. Campbell & Yogo (2006)
  - instead of treating its estimate as known.
- It combines a confidence set for persistence with tests of the predictive slope over that set.
  - allocating the error probability across the steps.
- For the S&P 500, $\hat{\rho} = 0.9988$ and $T(1 - \hat{\rho}) = 1.4$.
  - a unit-root test cannot reject $\rho = 1$.
  - there the $t$-statistic of $\hat{\beta}$ departs most from $N(0, 1)$.
- This Q-test concerns predictive regressions
  - it is not the Ljung–Box Q statistic for autocorrelations.

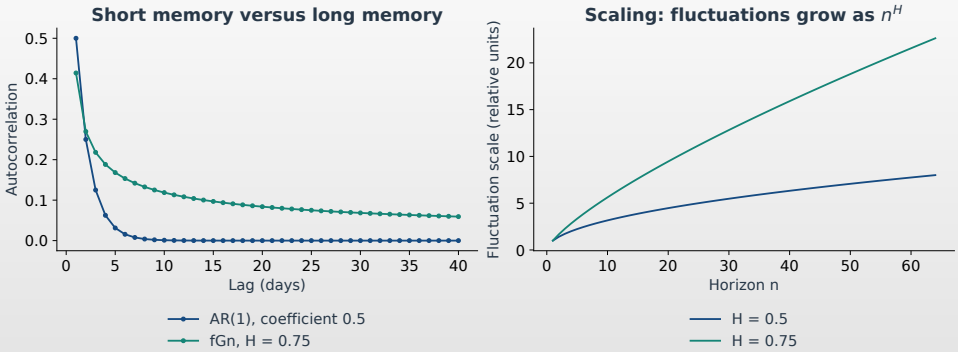
# Worked Example: the Stambaugh Bias for the S&P 500

- Monthly log excess return regressed on the lagged $dp_t$, 1934–2026, $T = 1\,111$ months
  - data construction: Appendix
- OLS: $\hat{\beta} = 0.267$, Newey–West $t = 0.91$: not significant even before any correction
  - in % per month per unit of $dp$
- Inputs: $\hat{\rho} = 0.9988$, $\text{corr}(\hat{u}, \hat{v}) = -0.97$, $\hat{\phi} = \hat{\sigma}_{uv}/\hat{\sigma}_v^2 = -0.925$ (as fractions, not in %)
- Substitution: bias $= -100\,\hat{\phi}\,(1 + 3\hat{\rho})/T = 0.333$ (in %, like $\hat{\beta}$)
  - corrected slope $= 0.267 - 0.333 = -0.066$: the estimated first-order bias correction exceeds the estimated slope
  - the correction is itself estimated: formal inference by Campbell & Yogo (2006) or a bootstrap
  - Kendall's formula is first order in $1/T$ and least accurate when $T(1 - \hat{\rho})$ is small, here 1.4: read $-0.066$ as indicative of the sign and size, not as a precise slope
- Simulation: the nominal 5% $t$-test rejects $\beta = 0$ in 26.5% of samples
  - with the same $\rho$ and $\text{corr}(u, v)$, $\beta = 0$ and a 30-year sample ($T = 360$ months)
  - more than five times too often

## Comparing Forecasts: Diebold–Mariano and Clark–West

- Section 3 defined $R^2_{OS}$ and applied the Clark–West test to the S&P 500
  - here we show why the test has this form
- Most published predictors have $R^2_{OS} < 0$ Welch & Goyal (2008)
  - the update Goyal, Welch & Zafirov (2024) confirms it
- The Diebold & Mariano (1995) test compares the mean difference between the squared errors of the two forecasts
- The *dp* model contains the historical mean (nested models)
  - under $H_0$ it estimates a zero $\beta$
  - the noise of that estimate inflates its mean squared prediction error (MSPE)
- So the Diebold–Mariano test rejects less often than its nominal level
- Clark & West (2007) add this noise back: $f_t = (r_t - \bar{r}_t)^2 - \left[(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2\right]$
  - $(\bar{r}_t - \hat{r}_t)^2$: the part of the model's error due to estimating $\beta$
  - the $t$ test on the mean of $f_t$ is one-sided: only a positive gain is evidence for the predictor

## Dependence across lags and horizons

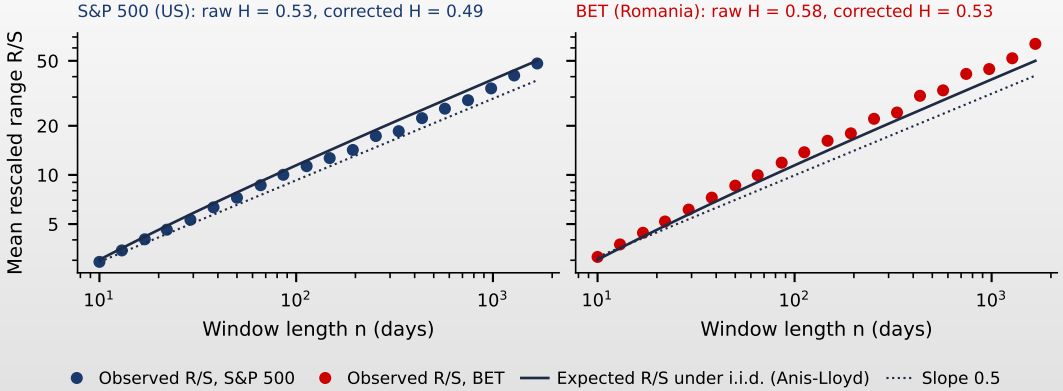



AR(1) correlations fade geometrically; the fGn example fades slowly. The right panel compares fluctuation growth across horizons.

## The Rescaled Range (R/S) Statistic

- Using the rescaled range, Hurst (1951) found that years of high Nile floods followed other high-flood years
  - more often than chance allows
  - history: Appendix
- Construction for a window of $n$ returns with mean $\bar{r}_n$ and standard deviation $s_n$
  - Cumulative deviations $Y_k = \sum_{i=1}^{k}(r_i - \bar{r}_n)$, $k = 1, \ldots, n$
  - Range $R_n = \max_k Y_k - \min_k Y_k$; rescaled range $R_n/s_n$
- Scaling law: $\mathbb{E}[R_n/s_n] \approx c\, n^H$, where $H$ is the **Hurst exponent**
  - $c$: a constant; taking logs, $\ln(R/S) \approx \ln c + H \ln n$, so $H$ is a slope
  - Estimate: slope of $\ln(R/S)$ on $\ln n$, averaging $R/S$ over non-overlapping windows
- $H = 0.5$: no long memory (short-run dependence allowed)
  - $H > 0.5$: persistence, long memory with $\rho_k \sim k^{2H-2}$
  - $H < 0.5$: anti-persistence

# R/S Analysis: S&P 500 and BET (1/2)




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# R/S Analysis: S&P 500 and BET (2/2)

- On the previous chart, the dots show the average $R/S$ for each window length $n$.
- The solid line is the Anis & Lloyd (1976) expectation for i.i.d. returns from the Normal distribution.
  - formula: Appendix.
- The Anis–Lloyd correction removes the bias of the raw slope.
  - The raw slope exceeds 0.5 in small samples even for i.i.d. returns.
  - $H_{AL} = 0.5+$ the slope of the regression of $\ln(R/S) - \ln \mathbb{E}[R_n/s_n]$ on $\ln n$.
- For the S&P 500, the raw exponent is 0.530 and the corrected one 0.485.
  - the apparent "persistence" came from small-sample bias.
- For the BET, the raw exponent is 0.578 and the corrected one 0.533.
  - persistence remains after the correction.
- Open question: is BET's $H > 0.5$ long memory, or just its lag-1 autocorrelation $\hat{\rho}_1 = 0.109$?
- The classical R/S statistic cannot tell the two apart.

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## Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Modification: $s_n$ is replaced by a long-run (HAC) standard deviation $\hat{\sigma}_\ell$ with $\ell$ lags
  - $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2\sum_{j=1}^{\ell}\left(1 - \frac{j}{\ell+1}\right)\hat{\gamma}_j$ , $\hat{\gamma}_j$: autocovariance at lag $j$ ($\hat{\gamma}_0$: the variance), with declining weights
  - $V = \dfrac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$ ; $Y_k$: cumulative deviations over the full sample of $T$ returns; bandwidth $\ell$ chosen as in Andrews (1991)
- Under short memory, $V$ lies in $[0.809, 1.862]$ with probability 95%
- $V$ is inside the interval: short memory is not rejected
  - for the S&P 500 ($V = 1.560$), Bitcoin (1.547), PX (1.749) and EUR/RON (1.429)
- For the BET ($V = 1.961$, with $\ell = 7$ lags) and Ethereum ($V = 1.980$), $V$ is just above 1.862
- This is weak evidence of long memory

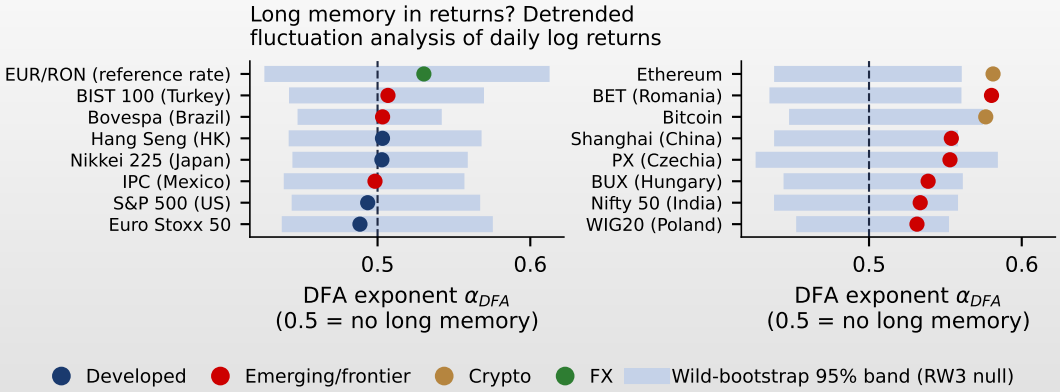
# Detrended Fluctuation Analysis (DFA) (1/2)

- Peng et al. (1994): DFA measures how the fluctuation of cumulative returns grows with the window length.
- Removing a linear trend from the profile removes a constant mean of returns in each box.
- If the mean of returns itself trends, higher-order DFA is needed.
- The algorithm
  - The profile $y_k = \sum_{t \leq k} (r_t - \bar{r})$ is the running sum of deviations from the mean, over the full sample.
  - The profile is split into boxes of length $n$.
  - In each box a linear trend is fitted and the residuals are kept.
  - $F(n)$ is the root mean square of all residuals; the calculation is repeated for many $n$.
- $F(n) \propto n^{\alpha_{DFA}}$: the DFA exponent $\alpha_{DFA}$ plays the role of $H$ ($\alpha_{DFA} = 0.5$: no long memory).

## Detrended Fluctuation Analysis (DFA) (2/2)

- The null band comes from a **wild bootstrap** (199 replications).
  - each residual $e_t = r_t - \bar{r}$ is multiplied by a random sign $\eta_t = \pm 1$, with probability $1/2$.
- The sign flip keeps the volatility $|e_t|$, allowed under RW3, and destroys only serial correlation.
  - a shuffle would destroy both.
- The implicit null is stricter than RW3: signs symmetric and independent, given volatility.
- The band is therefore a benchmark, not an exact test of RW3 and not a confidence interval for $\alpha_{DFA}$.

# DFA Exponent Across Markets (1/2)



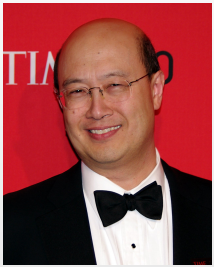

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# DFA Exponent Across Markets (2/2)

- Each dot on the previous chart is the exponent $\alpha_{DFA}$ of one series.
  - the bar is the 95% wild-bootstrap null band.
- S&P 500 (0.494), Euro Stoxx 50 (0.488) and Nikkei (0.503) are close to 0.5, inside the null band.
- Three of the 16 series exceed the null band.
  - Ethereum (0.581).
  - BET (0.580, against an upper limit of 0.561).
  - and, marginally, Bitcoin (0.576 against 0.575).
- PX (0.553) and Shanghai (0.554) exceed only a null band built by shuffling.
  - Shuffling also destroys volatility clustering.
  - their apparent memory comes from volatility clustering.
- The local Whittle estimator Robinson (1995) estimates $d = H - 0.5$ from the lowest frequencies.
  - for the BET, $\hat{d} = 0.118$, 95% interval $[0.062, 0.174]$ (Appendix).
- The interval assumes conditionally homoskedastic innovations.
  - volatility clustering violates this assumption.
  - so it is not decisive evidence.
- Conclusion: the evidence of long memory in returns is weak and depends on the method.
  - memory in volatility (Chapter 1) is strong.

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# Adaptive Markets Hypothesis (AMH) (1/2)



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0; Wikimedia Commons

- Full-sample tests give weak and unstable evidence
  - one possible reason: efficiency varies over time
- Andrew Lo proposed the Adaptive Markets Hypothesis (AMH) in Lo (2004) and developed it in the book Lo (2017)
- Markets are an ecology of competing species that adapt by trial and error
  - traders, funds, algorithms
  - not by perfect optimisation
  - efficiency is a matter of degree and depends on the number of competitors, the profit opportunities and the environment
- Anomalies appear, weaken and may reappear
- Time-varying statistics alone do not identify the AMH
  - since an Efficient Market Hypothesis (EMH) with time-varying risk premia also allows them
  - comparison table: Appendix

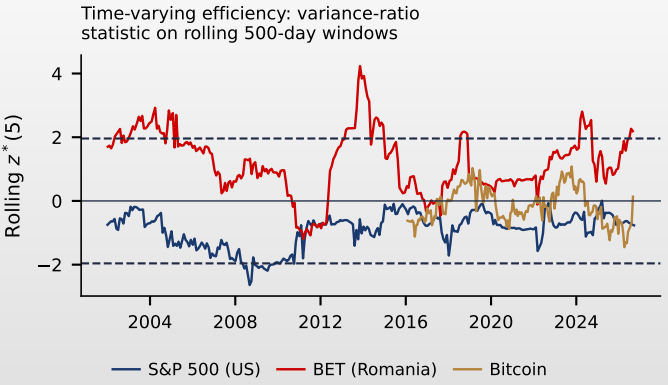
## Adaptive Markets Hypothesis (AMH) (2/2)

- Lim & Brooks (2011), a survey: stock-market efficiency varies over time, especially around crises
- Urquhart & McGroarty (2016): serial dependence appears and disappears across subperiods with market conditions
  - on four major indices (US, UK, Japan, euro area)

## Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**
  - 500 trading days (about two years)
  - moved forward by 21 days (one month)
  - Statistics: $z^*(5)$ of the variance-ratio test and the DFA exponent
- The DFA band is computed as before (random sign flips), but separately in each window
  - so it follows the window's volatility
- For comparison, a fixed null band is $[0.380, 0.615]$
  - simulated from i.i.d. Student-$t$ returns with 4 degrees of freedom
  - heavy tails, no volatility clustering
- The fixed band ignores volatility clustering, allowed under RW3
  - volatility clustering raises $\alpha_{DFA}$
  - so the fixed band flags too many windows
- Two cautions for reading the results
  - Overlapping windows are strongly correlated: runs of rejections are not independent evidence
  - Even under the null about 5% of windows reject; a share near 5% is a description, not a calibrated test of stability

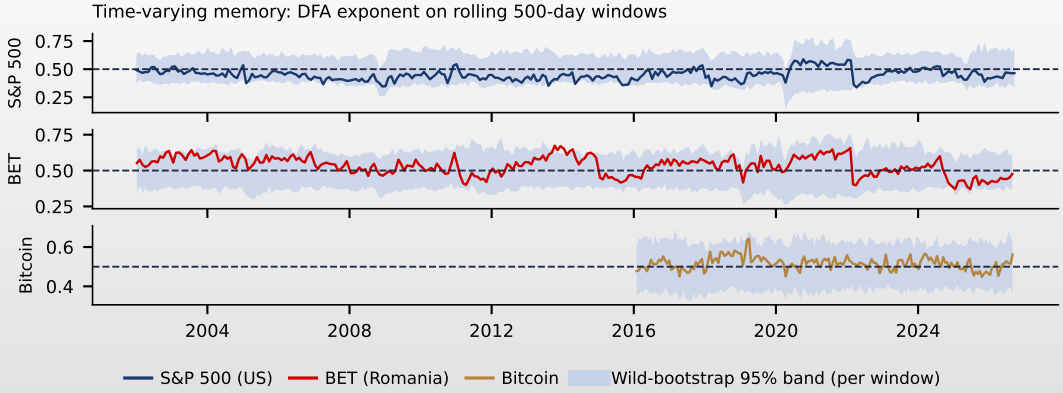
# Rolling Variance Ratio: S&P 500, BET, Bitcoin




- $z^*(5)$ on rolling 500-day windows, dated by the last day
  - dashed lines $\pm 1.96$
- Share of windows with $|z^*(5)| > 1.96$: S&P 500 6.7%, BET 22.4%, Bitcoin 0.0%
- S&P 500: negative in almost every window, beyond $-1.96$ only in windows ending in 2008–2010
- BET: episodes (windows ending in 2002–2005, 2013–2014, 2018 and after 2023)
- The FTSE upgrade of 21 Sep 2020 shows no break
  - BET was near zero from 2016 and rises again after 2023

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# Rolling DFA Exponent: S&P 500, BET, Bitcoin (1/2)




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## Rolling DFA Exponent: S&P 500, BET, Bitcoin (2/2)

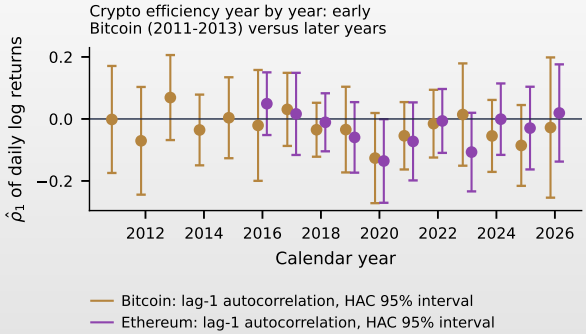
- $\alpha_{DFA}$ on rolling 500-day windows.
  - shaded: the wild-bootstrap 95% null band computed in each window.
- Outside their own band: S&P 500 1.7%, BET 3.1%, Bitcoin 0.0% of windows.
- With a fixed Student-$t_4$ band: 6.7%, 11.2%, 1.1%.
  - the fixed band ignores volatility clustering.
  - the extra DFA exceedances of the BET came from volatility clustering.
- The BET episodes in the variance-ratio chart remain.
  - they concern autocorrelation over a few days (short memory).
  - the DFA exponent does not target short memory.
- Low power: the BET band averages $[0.36, 0.64]$.
  - 500 observations cannot pin down a scaling exponent.

MFM_ch2_rolling_efficiency

# The AMH, Tested

- Kim, Shamsuddin & Lim (2011) show that predictability varies with market conditions (crises, bubbles).
  - rather than along a declining trend.
  - over a century of US data.
  - and make the AMH testable by linking it to state variables.
- Our version: regress the rolling $z^*(5)$ on the window's annualised volatility (%) and a crisis indicator
  - The crisis indicator is 1 if the window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020.
  - Newey–West errors with 24 lags, because consecutive windows share 95% of their days.
  - Volatility is the control variable: windows with similar volatility are compared.
- S&P 500: the crisis coefficient is $-0.71$ ($t = -2.53$).
  - the volatility coefficient 0.008 ($t = 0.39$).
  - $R^2 = 0.21$.
- BET: the crisis coefficient is $-0.82$ ($t = -2.77$).
  - the volatility coefficient $-0.008$ ($t = -0.36$).
  - $R^2 = 0.15$.
- In crisis windows $z^*(5)$ is lower, i.e. closer to reversal, even at equal volatility.
- What the result supports: association.
  - conditional on volatility, not a causal effect.
- State dependence is consistent with the AMH but does not identify it.
  - a change in $z^*(5)$ can come from $\widehat{\mathrm{VR}}(5) - 1$ or from its standard error.

# Crypto Efficiency Year by Year (1/2)




- The AMH predicts that a new market becomes more efficient over time
  - $\hat{\rho}_1 \neq 0$ in Bitcoin's early years, values near zero later
- The chart: $\hat{\rho}_1$ of daily returns in each calendar year, with the HAC 95% interval, for comparing years
- Bitcoin 2011–2013: the intervals are wide and include zero
  - the data do not show the early inefficiency the AMH predicts

MFM_ch2_crypto_efficiency

# Crypto Efficiency Year by Year (2/2)

- Only one of the 27 yearly autocorrelations is significant at 5%: Ethereum in 2023, $\hat{\rho}_1 = -0.107$
  - 16 for Bitcoin, 11 for Ethereum
  - by chance alone we would expect about 1.35
  - the test uses $\mathrm{SE}_0$ (derived under $H_0$), the chart's bars the HAC SE: the bar of Ethereum 2020 excludes 0, that of 2023 does not
- Urquhart (2016) rejects efficiency in the early years (Appendix)
  - with other tests, another price source and classical standard errors
- Power: one year of 365 returns gives a robust SE of about 0.05–0.13, so small effects are invisible
  - the robust SE requires a finite fourth moment, which Chapter 1 questions

## Event Studies: the Prediction Error

- Prediction-error variance MacKinlay (1997): $\mathrm{Var}(AR_\tau) = \sigma_\varepsilon^2\left[1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2}\right]$, $T_0$: estimation days
  - $\sigma_\varepsilon^2$: the residual variance of the market model; $r_{m,\tau}$: the market return on event day $\tau$.
  - $\hat{\mu}_m$, $\hat{\sigma}_m^2$: mean and variance of the market return in the estimation window.
  - The bracket grows when the event-day market move is unusual.
- For the bank tax (section 4), the extreme BET move makes the bracket 2.12.
  - hence $t = -6.85$ instead of $-10.0$ for TLV.
- With the same correction, BRD has $t = -4.03$.
- Boehmer, Musumeci & Poulsen (1991) standardise each $AR$ by its own prediction error.
  - estimate the variance from the firms' standardised values on the event day, not from calm days.
  - the test stays valid if the event raises the variance.

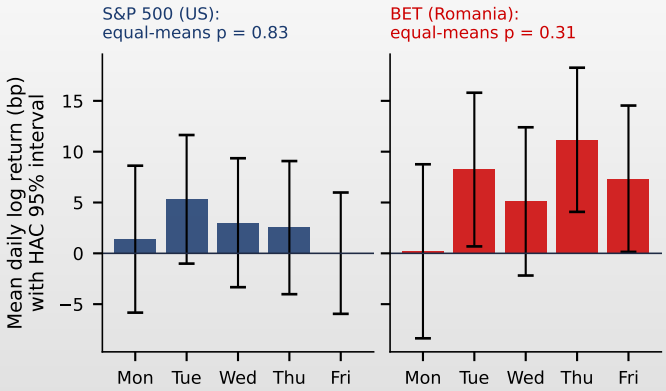
## Event Studies: Correlation Across Firms

- The question: what happens to the test of $\overline{AR}_0$ when all firms have the event on the same day?
- Let $N$ be the number of firms and $\bar{\varrho}$ the average correlation of their abnormal returns.
  - estimated in the estimation window.
  - for TLV and BRD, $\bar{\varrho} = 0.153$.
- The test that assumes independence ignores the covariances.
  - with $\bar{\varrho} > 0$ the true variance of the mean is $1 + (N-1)\bar{\varrho}$ times larger.
- For TLV and BRD, the $t$ computed under independence is $-7.55$.
  - divided by $\sqrt{1+\bar{\varrho}} = 1.074$ it becomes $-7.03$.
- That is the $t$ of the equal-weighted portfolio, which allows for the correlation automatically.
- If the variance is estimated cross-sectionally, as in Boehmer, Musumeci & Poulsen (1991), it further understates the true variance.
  - by the factor $1 - \bar{\varrho}$.
  - this is why Kolari & Pynnönen (2010) multiply the statistic by $\sqrt{(1-\bar{\varrho})/(1+(N-1)\bar{\varrho})}$.
- With ten firms and the same correlation, the independence-based test understates the variance.
  - By a factor of $1 + 9\bar{\varrho} = 2.38$.
  - the cross-sectional test by $(1+9\bar{\varrho})/(1-\bar{\varrho}) = 2.81$.

## Anomalies and Calendar Effects

- The question: which patterns survive the correction for multiple testing, and what explains them?
- An **anomaly** is a return pattern the asset-pricing model does not explain.
- Three possible explanations.
  - An omitted risk factor.
  - Mispricing (a behavioural bias).
  - Data snooping.
- Families:
  - calendar (section 5): TOM Lakonishok & Smidt (1988), first half of the month Ariel (1987).
  - time-series (time-series momentum).
  - cross-sectional (size, value, momentum: **Chapter 3**).
- Returns on published anomalies fall McLean & Pontiff (2016):
  - by 26% after the end of the sample studied.
  - by 58% after publication.
- Day of week: $r_t = \sum_{d=1}^{5} \beta_d D_{d,t} + u_t$.
  - $D_{d,t} = 1$ if day $t$ is weekday $d$ (Monday, ..., Friday), else 0.
  - $\beta_d$ is the mean return on weekday $d$.
- The Wald test of $\beta_1 = \cdots = \beta_5$ ($\chi^2_4$) uses HAC errors Newey & West (1987) with 5 lags.
  - because $u_t$ is heteroskedastic and autocorrelated.

# Day-of-the-Week Effect: S&P 500 and BET



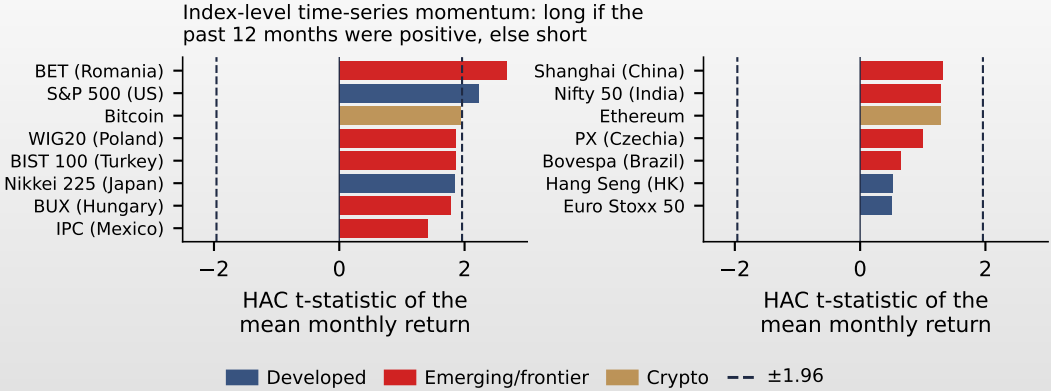

- Mean daily log return by weekday, in basis points (bp), 2000–2026, HAC 95% intervals
  - 1 bp = 0.01%
- S&P 500: we find no Monday effect since 2000, unlike French (1980)
  - from 0.0 bp (Friday) to 5.3 bp (Tuesday)
  - equal-means $p = 0.83$
- BET: Thursday 11.2 bp ($t = 3.09$), yet the joint test gives $p = 0.31$
- Two different nulls:
  - a day's mean equal to zero (mixes the day effect with the positive average return)
  - all five means equal (the day effect itself)

MFM_ch2_calendar_anomalies

## Time-Series Momentum on Indices (1/3)

- Time-series momentum (TSMOM) Moskowitz, Ooi & Pedersen (2012): signal $s_t = \text{sign}\left(\sum_{j=1}^{12} r_{t-j}\right)$
  - $+1$ (long) if the past 12 monthly log returns sum to a gain, $-1$ (short) otherwise
  - The strategy return is $s_t R_t$, $R_t$ the index return of month $t$
- On the next chart, each bar is the HAC $t$ (6 lags) of the strategy's mean monthly return in one market.

# Time-Series Momentum on Indices (2/3)




MFM_ch2_tsmom

# Time-Series Momentum on Indices (3/3)

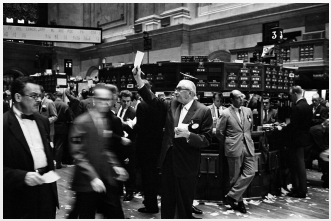
- Taken alone, the strategy has a mean return significant at 5%:
  - for the BET (1.40% a month, $t = 2.68$).
  - the S&P 500 ($t = 2.22$).
- For Bitcoin, $t = 1.94$, below 1.96.
- But we tested 15 markets.
  - the first Holm threshold is $0.05/15 = 0.0033$.
  - the smallest p-value, the BET's, is 0.007.
- After the multiple-testing correction, no market remains significant.
- The TSMOM Sharpe ratio is below buy-and-hold: 0.65 vs 0.75 for the BET, 0.48 vs 0.52 for the S&P 500.
- What the result supports: comparison of procedures.
- Part of the TSMOM return may be the equity premium, earned in the months spent long.
  - the mean stock return over the risk-free rate.
- The Sharpe ratio is annualised (full table: Appendix).
  - $\sqrt{12}\times$ monthly mean over monthly SD.
  - on price indices without dividends or a risk-free rate.

MFM_ch2_tsmom

## Behavioural concepts

- Prospect theory: gains and losses are judged relative to a reference point.
  - losses weigh more than equal gains (loss aversion).
  - small probabilities are overweighted.
- Disposition effect: the tendency to sell winners and keep losers.
- Anchoring: excess weight on an initial number, such as the purchase price.
- Overconfidence: overstating the precision of one's beliefs, which leads to too much trading.
- Limited attention: relevant information left unprocessed.
- Herding: following others' actions without an independent assessment.
- Underreaction and overreaction: a price adjusts too little or too much to news at first.
  - the second is followed by reversal.

# Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain; Wikimedia Commons

- **Excess volatility** Shiller (1981): prices move far more than the dividends that should justify them
- **Overreaction**: De Bondt & Thaler (1985) use it to explain 3–5-year return reversals
  - portraits: Appendix
- **Prospect theory** Kahneman & Tversky (1979) explains the disposition effect
- **Herding** in an illiquid market (BVB): herding produces log-periodic bubbles Pele, Mazurencu-Marinescu & Nijkamp (2013)
  - log-periodic bubble: prices rising super-exponentially, with ever faster oscillations before the crash
  - today its channels are order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency.
  - rational arbitrageurs could trade against them.
  - Can they?

# Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0; Wikimedia Commons

- Shleifer & Vishny (1997): real arbitrage is risky and needs capital
- **Noise-trader risk** De Long, Shleifer, Summers & Waldmann (1990): mispricing can get worse before it corrects
  - Noise traders: investors who trade on signals without information content (sentiment, rumours)
- Funding constraints: managers lose clients after losses and must close positions at the worst time
- Costs: short-selling limits, margins, trading costs, no perfect substitutes
- Anomalies can persist where arbitrage is hard
  - in small, illiquid markets such as the Romanian one (BET) we find the strongest persistence among the stock indices
- The AMH reconciles both views: rational and biased behaviour coexist and evolve

## Can Machine Learning Beat Efficient Markets?

- ML (machine learning) learns a prediction rule from data (Chapter 13)
  - captures nonlinear patterns that autocorrelation tests miss
- Gu, Kelly & Xiu (2020): trees and neural networks give large economic gains over linear regressions
  - in the US cross-section
  - yet their monthly out-of-sample $R^2$ is below 1%
- Large language models read news and filings: Tak & Pele (2025) forecast stock trends with LLaMA

## Validation, leakage and shrinkage

- Flexible models, such as trees and neural networks, can also fit accidental sample patterns (overfitting).
- Walk-forward validation: the model is estimated only on data before the test period.
- Leakage: a forecast uses information unavailable at its date.
  - such as an accounting figure revised later.
- Ridge: $\hat{g} = \arg\min_g \sum_t (r_t - x_t' g)^2 + \lambda \sum_j g_j^2 = (X'X + \lambda I)^{-1} X' r$.
  - $\lambda > 0$ pulls the coefficients towards zero (shrinkage).
  - reduces estimation variance.
- With $N(0, \sigma^2)$ errors and a prior $g \sim N(0, \tau^2 I)$, ridge with $\lambda = \sigma^2/\tau^2$ is the posterior mean of the coefficients.
  - Bayesian learning.
- Cross-validation: leave out one block of data at a time (e.g. a year).
  - estimate on the rest.
  - pick the $\lambda$ with the smallest error on the blocks left out.

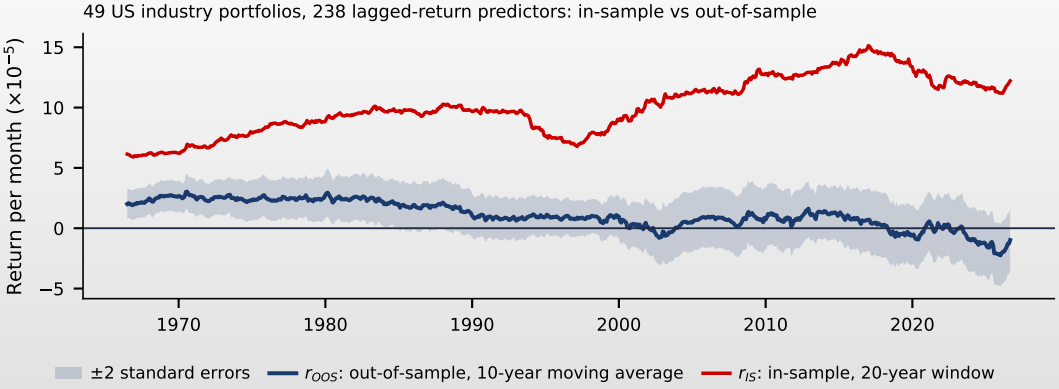
# Case Study: Rational Learning with Many Predictors (1/2)

- Martin & Nagel (2022): investors price $N$ assets whose cash flows depend on $J$ characteristics with **unknown** coefficients
  - learn them optimally (Bayes) from past data
  - research question: Appendix
  - The posterior mean of the coefficients is the ridge estimator of the previous slide.
  - Investors are risk-neutral, so predictability cannot be a risk premium.
- **In sample**: the test of no predictability rejects with probability tending to 1 (Martin & Nagel (2022), Prop. 4).
  - if the number of predictors per asset, $J/N$, tends to a constant $\psi > 0$.
- The mechanism: the price embeds the shrunk coefficients, so it underreacts to $X$.
  - ex post, $X$ predicts returns even though investors are rational.
- A rejection is not evidence of irrationality (formal results: Appendix).
  - the rational-expectations null is the wrong benchmark when investors must learn the parameters.

# Case Study: Rational Learning with Many Predictors (2/2)

- $r_{OOS}$: the return of a portfolio built from forecasts.
  - in a period not used to estimate its coefficients.
- **Out of sample**: mean return zero (Martin & Nagel (2022), Prop. 6).
  - For a portfolio built with coefficients from other periods.
  - $\mathbb{E}[r_{OOS}] = 0$.
  - model and empirical design: Appendix.
- So out-of-sample tests keep their economic meaning: in this model, they test efficiency.
- For US stocks, the in-sample $R^2$ is about 1%.
- The out-of-sample return $r_{OOS}$ is much lower than the in-sample one (Martin & Nagel (2022), Fig. 8).
  - after 2014, within two standard errors of zero.

# Replication on Course Data: In Sample vs Out of Sample (1/2)



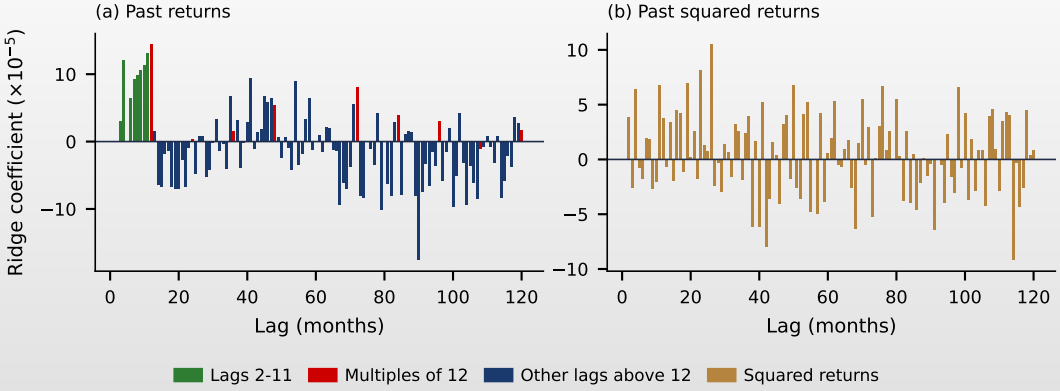

MFM_ch2_big_data

# Replication on Course Data: In Sample vs Out of Sample (2/2)

- Data: 40–49 industry portfolios (Kenneth French Data Library), July 1926 – August 2026.
  - 49 in recent years.
  - 238 predictors (lagged returns and their squares, lags 2–120).
  - rolling 20-year OLS, as in Martin & Nagel (2022), Sec. 7.
- The portfolio: each month, every industry gets a weight proportional to its forecast return.
- Vertical axis: the portfolio's monthly return, in units of $10^{-5}$.
  - one series is $r_{\text{IS}}$ (coefficients from the same window).
  - the other the 10-year moving average of $r_{\text{OOS}}$ (coefficients from the previous window), with a descriptive $\pm 2$ standard-error band.
- $r_{\text{IS}}$ is between 5.90 and 15.14, positive by construction.
- $r_{\text{OOS}}$ has a mean of $1.05 \times 10^{-5}$ over 842 months (from July 1956), about 11% of $r_{\text{IS}}$.
- In the first 30 out-of-sample years (to June 1986), the mean of $r_{\text{OOS}}$ is 4.16 standard errors above zero, afterwards only 0.62.
- With $J/N \approx 5$, the in-sample fit is inflated: an $R^2$ of 5.4% against about 1% in Martin & Nagel (2022) (Appendix).
- What the result supports: comparison of procedures.

MFM_ch2_big_data

# Replication on Course Data: Ridge Coefficients by Lag (1/2)




MFM_ch2_big_data

## Replication on Course Data: Ridge Coefficients by Lag (2/2)

- Full-sample ridge, January 1971 – August 2026.
  - the penalty $\lambda$ is chosen by leave-one-year-out cross-validation (56 years).
- The previous chart: the ridge coefficient of each standardised predictor, by lag in months.
  - compare Martin & Nagel (2022), Fig. 7.
- For lags 2–11, 8 of 10 coefficients are positive, matching **momentum** Jegadeesh & Titman (1993).
- For multiples of 12, 9 of 10 coefficients are positive, matching **seasonality** Heston & Sadka (2008).
- For the other lags above 12, 67% of the coefficients are negative, matching long-run **reversal** De Bondt & Thaler (1985).
  - the coefficients on squared returns have no dominant sign.
- $R^2$ on the years left out, $1 - \sum(r - \hat{r})^2 / \sum r^2$: ridge 0.013%, OLS −2.26%.
- Ridge recovers the known anomalies (limits and extensions: Appendix).
  - yet they predict almost nothing in the years left out.

MFM_ch2_big_data

## ETFs, placebo and price discovery

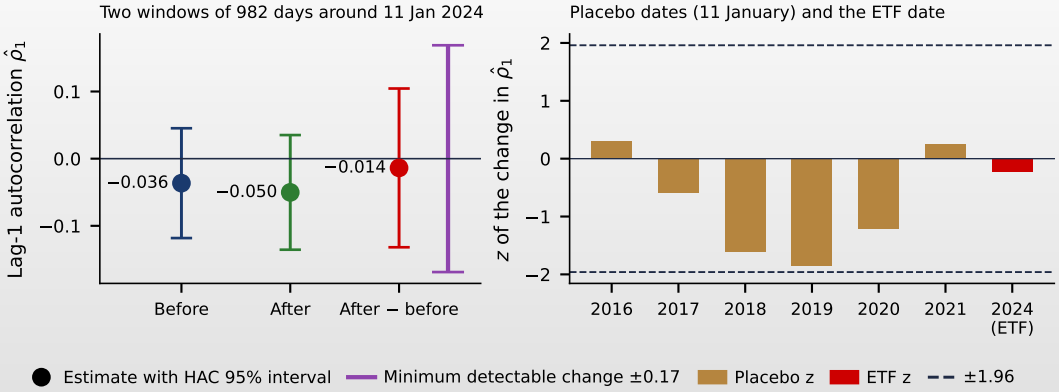
**ETF**: Exchange-Traded Fund.

- A spot ETF holds the underlying asset (here Bitcoin).
  - not futures contracts like the Bitcoin ETFs launched in 2021.
  - arbitrage between the fund and the spot market becomes direct.
- A placebo repeats the analysis at a date without the event.
  - a similar apparent change in 2019 weakens a claim that the 2024 launch caused the change.
- Price discovery is the incorporation of information into prices.
  - if exchange A reacts to news first and B follows, A may contribute more to price discovery.

## Estimation and Power (1/3)

- The test: two non-overlapping windows of 982 days, before and after 11 January 2024.
- $z = \Delta\hat{\rho}_1 / \sqrt{s^2_{\text{pre}} + s^2_{\text{post}}}$, with $\Delta\hat{\rho}_1 = \hat{\rho}_{1,\text{post}} - \hat{\rho}_{1,\text{pre}}$ and the HAC standard errors $s$ of the two estimates (valid also when $\rho_1 \neq 0$).
- **Minimum detectable effect (MDE)**: the smallest true change detected with 80% power at the 5% level.
- $\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE}(\Delta\hat{\rho}_1) \approx 2.8 \times \text{SE}(\Delta\hat{\rho}_1)$
  - $z_p$: the $p$-quantile of the standard Normal distribution; $z_{0.975} = 1.96$ and $z_{0.80} = 0.84$
- Placebo dates: 11 January of each year 2016–2021, with windows of 982 returns.

# Mini-Case: Estimate, Placebo and MDE




MFM_ch2_ai_discovery

## Estimation and Power (2/3)

- The previous chart, left: $\hat{\rho}_1$ before and after the launch, with HAC 95% intervals.
  - the third point is the difference, with the $\pm$MDE bar.
- Right: the $z$ statistic for the ETF date, among those of the six placebo dates.
- Before (May 2021 – Jan 2024): $\hat{\rho}_1 = -0.036$ ($s = 0.042$), VR(5) $= 1.000$ ($z^* = 0.00$).
- After (Jan 2024 – Sep 2026): $\hat{\rho}_1 = -0.050$ ($s = 0.044$), VR(5) $= 0.939$ ($z^* = -0.64$).
- $\Delta\hat{\rho}_1 = -0.014$, $z = -0.23$: an unchanged $\rho_1$ is not rejected.
  - The HAC standard errors $s$ require a finite fourth moment, which Chapter 1 questions.
- None of the six placebo dates gives a significant change in $\rho_1$ at 5%.
  - the largest statistic is $|z| = 1.85$.
- What the result supports: description.
  - A before/after comparison without a control group (difference-in-differences).

## Estimation and Power (3/3)

- Here $\mathrm{SE}(\Delta\hat{\rho}_1) = \sqrt{0.042^2 + 0.044^2} \approx 0.060$, so the MDE is about 0.17.
- "Not significant" is not "no effect": the MDE is about 12 times the estimated change.
- Before 2024, $\hat{\rho}_1 = -0.036$.
  - even a full move to $\rho_1 = 0$ would be a change almost five times smaller than the MDE.
  - this design would detect it with probability well below 80%.
- A better outcome variable: intraday price discovery across venues, not daily $\rho_1$.

MFM_ch2_ai_discovery

## Difference-in-differences: example and assumptions

- The before/after test cannot separate the effect of the ETFs from other changes in 2024.
- A causal conclusion needs a control group, for example Ethereum before its own ETFs (July 2024).
- Difference-in-differences (DiD) compares two before–after changes.
  - It subtracts the control group's change from the treated group's change.
- Parallel trends:
  - without treatment, the treated and control groups would have changed in the same way.
- Similar pre-treatment trends support this assumption but cannot prove it.
- No anticipation: the treated group does not react before treatment.
- No spillover: treatment does not affect the control group.
  - for example, a Bitcoin ETF must not change Ethereum trading.

# Appendix: Pioneers, Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds move like a random walk
  - "The mathematical expectation of the speculator is zero"
- Samuelson (1965): if prices are **properly anticipated**, their changes are unpredictable
- Both results are the ancestors of the Efficient Market Hypothesis

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# Appendix: Eugene Fama



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Fama (1970) gave the definition used in this chapter and the three forms of efficiency
- Nobel Prize 2013, shared with Robert Shiller (a sceptic of efficiency) and Lars Peter Hansen
- Update and survey: Fama (1991)
  - an accessible defence of the EMH against its critics: Malkiel (2003)

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# Appendix: Summary Table, Weak-Form Tests 2000–2026

| Market | $T$ | $\hat{\rho}_1$ | $\mathrm{SE}_0$ | $\mathrm{SE}_{\mathrm{HAC}}$ | $\widehat{\mathrm{VR}}(5)$ | $z^*(5)$ | CD $p$ | Runs $z$ |
|---|---|---|---|---|---|---|---|---|
| S&P 500 (US) | 6 717 | −0.099 | 0.027 | 0.030 | 0.829 | −2.78 | 0.001 | 4.34 |
| Euro Stoxx 50 | 6 836 | −0.029 | 0.019 | 0.016 | 0.910 | −2.08 | 0.097 | 3.41 |
| Nikkei 225 (Japan) | 6 545 | −0.038 | 0.024 | 0.018 | 0.923 | −1.48 | 0.317 | 1.97 |
| Hang Seng (HK) | 6 583 | −0.004 | 0.024 | 0.017 | 0.967 | −0.64 | 0.872 | −0.17 |
| BET (Romania) | 6 670 | 0.109 | 0.028 | 0.023 | 1.173 | 2.98 | 0.001 | −6.05 |
| WIG20 (Poland) | 6 686 | 0.018 | 0.018 | 0.016 | 1.010 | 0.25 | 0.775 | 2.25 |
| BUX (Hungary) | 6 666 | 0.037 | 0.024 | 0.021 | 1.013 | 0.26 | 0.427 | −0.42 |
| PX (Czechia) | 6 697 | 0.051 | 0.032 | 0.020 | 1.041 | 0.60 | 0.354 | −2.53 |
| BIST 100 (Turkey) | 6 692 | 0.005 | 0.024 | 0.018 | 1.033 | 0.64 | 0.949 | −0.63 |
| Bovespa (Brazil) | 6 622 | −0.028 | 0.025 | 0.027 | 0.928 | −1.29 | 0.521 | 0.88 |
| IPC (Mexico) | 6 735 | 0.071 | 0.019 | 0.016 | 1.038 | 0.92 | 0.001 | −3.27 |
| Nifty 50 (India) | 6 613 | 0.048 | 0.024 | 0.019 | 1.048 | 0.91 | 0.178 | −5.46 |
| Shanghai (China) | 6 474 | 0.021 | 0.019 | 0.018 | 1.046 | 1.11 | 0.409 | 1.02 |
| Bitcoin | 4 384 | −0.022 | 0.026 | 0.021 | 0.991 | −0.19 | 0.835 | 2.68 |
| Ethereum | 3 913 | −0.012 | 0.027 | 0.023 | 1.067 | 1.17 | 0.027 | 3.91 |
| EUR/RON (reference rate) | 5 352 | 0.170 | 0.036 | 0.033 | 1.157 | 2.19 | 0.000 | −1.66 |

- $T$: number of daily returns
  - $\mathrm{SE}_0$: robust SE of $\hat{\rho}_1$ under $H_0$, for the test; $\mathrm{SE}_{\mathrm{HAC}}$: HAC SE of the estimator, for intervals
  - CD: Chow–Denning over $q = 2, 5, 10, 20$
  - EUR/RON from July 2005, crypto on 365 days per year

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# Appendix: Runs Test on All Markets, Other VR Tests

- Runs $z$ (table on the previous slide):
  - Nifty 50 $-5.46$, IPC $-3.27$ (persistence in emerging markets)
  - Ethereum 3.91, significant although $\hat{\rho}_1$ is not
- Choi (1999): an automatic VR test that chooses the horizon from the data
  - spectral estimate at frequency zero
- Wild-bootstrap versions for small samples: Kim (2006)
  - applications to Asian markets: Kim & Shamsuddin (2008)

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# Appendix: Automatic Portmanteau Test of the MDH

- What we derive: a portmanteau test valid under the MDH with conditional heteroskedasticity Escanciano & Lobato (2009).
  - the number of lags chosen from the data.
- $AQ = T\sum_{j\le\tilde{p}}\tilde{\rho}_j^2$, with $\tilde{\rho}_j = \hat{\gamma}_j/\sqrt{\hat{\kappa}_j}$, $\hat{\gamma}_j = \frac{1}{T-j}\sum_t e_t e_{t-j}$ and $\hat{\kappa}_j = \frac{1}{T-j}\sum_t e_t^2 e_{t-j}^2$
  - $e_t = r_t - \bar{r}$; $\hat{\gamma}_j$: the autocovariance at lag $j$; $\hat{\kappa}_j$: its robust variance
  - $\tilde{p} \le 10$ maximises $T\sum_{j\le p}\tilde{\rho}_j^2$ minus a penalty: BIC ($p \ln T$), but AIC ($2p$) when $\max_j \sqrt{T}|\tilde{\rho}_j|$ exceeds $\sqrt{2.4 \ln T}$
- Under the MDH, $\tilde{p} \to 1$ with probability 1, hence the $\chi_1^2$ limit.
  - a large $AQ$ rejects the MDH.
- $AQ$: S&P 500 13.23 ($p = 0.0003$), BET 14.92 ($p = 0.0001$), Bitcoin 0.75 ($p = 0.385$).
  - the MDH is rejected at 5% for 5 of the 16 series.

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# Appendix: Predictive Regression, Data and Full Output

**FRED**: Federal Reserve Economic Data.

- Monthly log excess return regressed on the lagged $dp_t$
  - dividends included, minus the 3-month T-bill rate
  - Data: Shiller (2026) (S&P Composite, to June 2026), extended with monthly averages of our daily S&P 500 closes (scaled over the last 12 common months) and the dividends of the SPDR S&P 500 ETF (SPY); 3-month US Treasury bill yield (FRED, series TB3MS; annual %), monthly log rate $\ln(1 + \text{TB3MS}/1200)$ lagged one month
- OLS: $\hat{\beta} = 0.267$, Newey–West SE 0.293 (12 lags), $t = 0.91$, OLS $t = 1.16$, $R^2 = 0.12\%$
- Corrected slope / unadjusted HAC SE $= -0.22$: only illustrative
  - it ignores the uncertainty of the correction
- Economic value Campbell & Thompson (2008): proportional rise in the expected return of the market timer
  - number: slide The Dividend–Price Ratio: Results for the S&P 500
  - $\dfrac{R^2}{1-R^2} \cdot \dfrac{1+S^2}{S^2}$ , $S$ the monthly Sharpe ratio
- The Diebold & Mariano (1995) test gives $t = -0.57$
  - but it favours the small model when the models are nested
  - Clark & West (2007) correct for this and give $t = 0.42$

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## Appendix: How Fast Is "Fast"?

**CME**: Chicago Mercantile Exchange.

- At high frequency, futures prices react to scheduled macro news within microseconds or less
  - Tak, Schlamp, Osterrieder & Pele (2026) measure this race at Eurex and CME
- Crypto events: Pele, Lin, Wang & Gaman (2025) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency
  - it depends on technology (Chapter 10, market microstructure)

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# Appendix: Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia Commons



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

- Hurst (1951) designed Nile reservoirs
- He found that the range of cumulative Nile flows grows faster than $\sqrt{n}$ with the horizon
  - years of high floods follow other years of high floods, hence the **rescaled range** statistic (R/S)

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# Appendix: The Anis–Lloyd Expectation

- Anis & Lloyd (1976): exact $\mathbb{E}[R_n/s_n]$ under i.i.d. returns from the Normal distribution
  - $\mathbb{E}[R_n/s_n] = \dfrac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\dfrac{n-i}{i}}$
  - $\Gamma$: the gamma function
- This $\mathbb{E}[R_n/s_n]$ is the reference subtracted in $H_{AL}$ (R/S Analysis slide)
- Uncorrected estimates overstate memory by about 0.04–0.05 with daily data

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# Appendix: Long Memory with Inference, the Local Whittle Estimator

- $d = H - 0.5$, with spectral density $f(\lambda) \sim G\lambda^{-2d}$ as $\lambda \to 0$
  - $G > 0$ a constant
  - $d > 0$: the density explodes at low frequencies, long memory
  - fit only the first $m$ Fourier frequencies $\lambda_j = 2\pi j/T$ ($m$: bandwidth)
  - $\hat{d} = \arg\min_d \left[ \ln\left(\frac{1}{m}\sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j)\right) - \frac{2d}{m}\sum_{j=1}^{m} \ln \lambda_j \right]$, $I$: the periodogram (sample spectral density)
  - Robinson (1995): for $|d| < 1/2$, $m \to \infty$, $m/T \to 0$, a smooth $f$ near 0 and a linear process with conditionally homoskedastic martingale-difference innovations, $\sqrt{m}(\hat{d} - d) \to N(0, 1/4)$ without Gaussianity: a confidence interval instead of a band
- Estimates with $m = T^{0.65}$ and the 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$
  - S&P 500 $-0.013$ $[-0.069, 0.043]$; Bitcoin 0.063 $[-0.002, 0.127]$
  - BET 0.118 $[0.062, 0.174]$; Ethereum 0.112 $[0.045, 0.179]$
- Bandwidth trade-off: a large $m$ reaches frequencies where short-run autocorrelation biases $\hat{d}$ upward
  - BET $\hat{\rho}_1 = 0.109$
  - with $m = T^{0.5}$ BET gives $\hat{d} = 0.135$ (SE 0.056): a sensitivity check supporting, not proving, that its $d > 0$ is not only short memory
  - GARCH returns also strain the conditions above

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# Appendix: EMH versus AMH

| | EMH | AMH |
|---|---|---|
| Investors | Rational, optimising | Bounded rationality, heuristics |
| Efficiency | An equilibrium property, tested as a restriction | A matter of degree that changes with conditions |
| Risk–return | Equilibrium risk premia, possibly time-varying | Changes with the population of investors |
| Anomalies | Omitted risk factors or data mining | Transient profit opportunities |
| Empirical tool | Tests of an equilibrium restriction | Tests of time and state variation (rolling windows) |

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# Appendix: Is Bitcoin Efficient? The Literature

- Urquhart (2016): Bitcoin inefficient over 2010–2016
  - some tests indicate efficiency in the later subperiod
  - His battery (Ljung–Box, runs, the rank test of Bartels (1982), automatic VR, BDS, R/S) includes BDS, which rejects under volatility clustering alone: not a test of RW3
- Nadarajah & Chu (2017): with a power transformation of returns, Bitcoin is weak-form efficient
- Kristoufek (2018): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- Tran & Leirvik (2020): time-varying efficiency across many crypto-assets, increasing over time
- Pele et al. (2023): crypto-assets have become a distinct asset class
  - their returns share the stylised facts of other markets
- Our own estimates, calendar year by calendar year
  - slide Crypto Efficiency Year by Year
  - Bitcoin, lag-1 autocorrelation $\hat{\rho}_1$: $-0.002$ in 2011, $-0.071$ in 2012, 0.069 in 2013, all close to zero
  - Robust variance-ratio statistic $z^*(5)$: at most 1.71 in absolute value over all crypto-asset years, below the critical value 1.96
  - In 2011–2013 the robust standard error of $\hat{\rho}_1$ is 0.09–0.11: the yearly non-rejection has little power and does not by itself contradict Urquhart (2016)

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# Appendix: Romania, Frontier then Emerging

**BVB**: Bursa de Valori București — the Bucharest Stock Exchange.

- FTSE Russell (2019): Romania reclassified from Frontier to Secondary Emerging
  - announced in September 2019
  - effective 21 September 2020
- Expected channels: more foreign investors, more analysts, more liquidity ⇒ faster price discovery
- Earlier BVB evidence:
  - crashes and log-periodic bubbles Pele & Mazurencu-Marinescu (2012)
  - herding in an illiquid market Pele, Mazurencu-Marinescu & Nijkamp (2013)
- Did efficiency improve? Seminar C2 compares five years before and after
  - with WIG20 and BUX as a comparison group
  - with a block bootstrap Künsch (1989)
  - A before/after comparison is descriptive unless confounders are controlled

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## Appendix: Reality Check, SPA and StepM explained

**SPA**: Superior Predictive Ability (test).

- Reality Check asks whether the best of all tried rules beats a benchmark. White (2000)
  - using a bootstrap distribution of their maximum performance.
- SPA standardises performance by its uncertainty (studentisation) and reduces distortion from poor rules. Hansen (2005)
- StepM tests the strongest statistic first, then the remaining family using jointly resampled critical values
  - it controls FWER under its assumptions. Romano & Wolf (2005)
- Example: the same months are drawn for all 15 calendar tests, so common market shocks are kept
  - separate resampling would destroy that dependence.

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# Appendix: StepM on the 15 Calendar Tests

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
  - Reality Check, SPA and StepM allow for it (explained on the previous slide)
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months
  - common to all markets
  - 999 replications
  - Critical $p$ at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
  - Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- Chordia, Goyal & Saretto (2020): thresholds $|t| > 3.8$ (time series) and $|t| > 3.4$ (cross-section)
  - from over two million strategies generated at random from real accounting data
  - without them, about 45% of rejections are false
- Lowering false discoveries raises missed ones Harvey & Liu (2020)

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# Appendix: Time-Series Momentum, All Markets

| Market | Mean (%/month) | $t_{HAC}$ | Sharpe TSMOM | Sharpe buy-and-hold | $\hat{\rho}_1$ monthly |
|---|---|---|---|---|---|
| S&P 500 (US) | 0.61 | 2.22 | 0.48 | 0.52 | 0.03 |
| Euro Stoxx 50 | 0.16 | 0.51 | 0.11 | 0.16 | 0.05 |
| Nikkei 225 (Japan) | 0.61 | 1.85 | 0.39 | 0.42 | 0.09 |
| Hang Seng (HK) | 0.18 | 0.53 | 0.10 | 0.19 | 0.04 |
| BET (Romania) | 1.40 | 2.68 | 0.65 | 0.75 | 0.19 |
| WIG20 (Poland) | 0.67 | 1.86 | 0.36 | 0.26 | 0.02 |
| BUX (Hungary) | 0.72 | 1.79 | 0.40 | 0.64 | 0.10 |
| PX (Czechia) | 0.38 | 0.99 | 0.24 | 0.45 | 0.10 |
| BIST 100 (Turkey) | 1.03 | 1.85 | 0.37 | 0.75 | −0.08 |
| Bovespa (Brazil) | 0.28 | 0.65 | 0.14 | 0.51 | 0.11 |
| IPC (Mexico) | 0.44 | 1.41 | 0.32 | 0.63 | 0.03 |
| Nifty 50 (India) | 0.50 | 1.29 | 0.28 | 0.64 | 0.02 |
| Shanghai (China) | 0.59 | 1.32 | 0.30 | 0.23 | 0.08 |
| Bitcoin | 4.47 | 1.94 | 0.72 | 1.08 | 0.13 |
| Ethereum | 6.50 | 1.28 | 0.57 | 0.90 | 0.16 |

- 307 months for equities, fewer for crypto
  - Sharpe ratio $\sqrt{12}\,\bar{x}/s_x$ of price returns, no risk-free rate subtracted
  - $\bar{x}$, $s_x$: mean and SD of the monthly strategy returns
  - Moskowitz, Ooi & Pedersen (2012) use futures excess returns scaled to 40% ex-ante volatility

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# Appendix: Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Richard H. Thaler, Nobel Prize 2017

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- Excess volatility Shiller (1981) and overreaction De Bondt & Thaler (1985): Psychology in Prices slide

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# Appendix: Case Study, Research Question

- Martin & Nagel (2022): what does a rejected no-predictability test mean?
  - when the number $J$ of characteristics is comparable to the number $N$ of assets
  - the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices "fully reflect" public information Fama (1970), with two readings
  - investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
  - investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the "factor zoo" Harvey, Liu & Zhu (2016)
  - text, satellite images and social media keep increasing $J$
  - does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: out-of-sample predictability Campbell & Thompson (2008) and the decay of anomalies after publication McLean & Pontiff (2016)

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# Appendix: Case Study, Model and Method (1/2)

- **Model** (Martin & Nagel (2022), Sec. 2)
  - cash-flow growth $\Delta y_t = Xg + e_t$, with $e_t \sim N(0, I)$ and a zero interest rate
  - $\Delta y_t$: the $N$ cash-flow growth rates; $X$: the $N \times J$ matrix of characteristics; $g$: the $J$ unknown coefficients; $I$: identity matrix
- **Learning** (Prop. 1–2)
  - prior $g \sim N(0, (\theta/J)\, I)$, the true distribution from which $g$ is drawn; $\theta$: signal strength
  - schematically, one period: the price is $X\hat{g}_t$, with $\hat{g}_t = (tX'X + (J/\theta)I)^{-1}X' \sum_{s \le t} \Delta y_s$, so $\mathbb{E}[\hat{g}_t \mid g] = Ag$, with $A$ "smaller" than $I$
  - $r_{t+1} = X(I - A)g$ (underreaction from shrinkage) $-X(\hat{g}_t - Ag)$ (estimation noise) $+e_{t+1}$ (unpredictable shock)
- **Asymptotics and simulations** (Sec. 4)
  - $N, J \to \infty$ with $J/N \to \psi > 0$
  - simulations with $N = 1000$, entries of $X$ drawn from the standard Normal distribution and $\theta = 1$

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# Appendix: Case Study, Model and Method (2/2)

- **CRSP**: Center for Research in Security Prices.
- **NYSE**: New York Stock Exchange.
- **Tests**
  - in sample: a cross-sectional OLS of $r_{t+1}$ on $X$ (requires $J < N$) with a joint $\chi^2$ test
  - out of sample: a portfolio with weights $\frac{1}{N} X h_{s+1}$, from coefficients estimated in other periods
  - $r_{\text{OOS},t+1} = \frac{1}{N} h'_{s+1} X' r_{t+1}$, $s \neq t$ (eq. 19); $r_{\text{IS}}$: the same with $s = t$
  - $h_{s+1}$: the estimated coefficient vector; $r_{t+1}$: the $N$ returns
- **Empirical design** (Sec. 7)
  - US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
  - predictors: simple and squared monthly returns at lags $t-2$ to $t-120$ (238 predictors), demeaned and standardised across stocks every month
  - full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation
  - rolling 20-year OLS gives $r_{\text{OOS}}$ from January 1956

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# Appendix: Case Study, Formal Results (1/2)

- In sample: the probability of no rejection is of the order $e^{-0.015N}$ (Fig. 2)
  - Prop. 4 of Martin & Nagel (2022), main slide
  - for $\psi > 0.4$ ($\theta = 1$, one period of learning)
- Simulations: a test of nominal size 5% rejects with probability above 90% once $J > N/2$ (Fig. 3b)
  - $N = 1000$, four periods of learning
  - not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with $k$-FWER control (Fig. 4)
  - probability of $k$ or more false rejections at most 5%
  - for $k = 20$ and $J/N = 0.05$, the probability of at least $k$ rejections is 25% instead of 5%
  - about 80% for $J/N$ close to 1
  - multiple-testing corrections do not help: the problem is the null hypothesis, not the test

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# Appendix: Case Study, Formal Results (2/2)

- Out of sample: $\mathbb{E}[r_{\text{OOS},t+1}] = 0$ holds forward and backward in time (Prop. 6)
  - positive only if investors shrink more than the data warrant (Fig. 6)
- US stocks: the ridge coefficients recover three patterns (Fig. 7)
  - momentum
  - long-run reversal
  - momentum seasonality
  - $r_{\text{IS}} \approx 0.006\%$ per month (Fig. 8)

# Appendix: Case Study, In Sample vs Out of Sample in Detail

- $r_{\text{IS}} \geq 0$ in every 20-year window by construction: $r_{\text{IS}} = b' G^{-1} b/240$
  - $G = \sum_s X_s' X_s / N_s$ and $b = \sum_s X_s' r_s / N_s$ over the 240 months $s$ of the window
  - $X_s$: the $N_s \times 238$ predictor matrix of month $s$; $r_s$: the $N_s$ portfolio returns; $G^{-1}b$: the pooled OLS coefficients
  - mean in-sample $R^2$ of 5.4%, against about 1% for individual stocks in Martin & Nagel (2022): 238 predictors against only 40–49 portfolios a month, pooled over 240 months, inflate the in-sample fit
- Out of sample (from July 1956):
  - until June 1986: mean $r_{\text{OOS}} = 1.98 \times 10^{-5}$, 4.16 standard errors above 0
  - afterwards $0.36 \times 10^{-5}$, only 0.62 standard errors
  - the 10-year average is last above two standard errors in the window ending in December 1987; in the windows ending in 2014–2019 its largest ratio to the standard error is 1.25, as for stocks
- Interpretation suggested by Martin & Nagel (2022): the gap is consistent with investor learning
  - but does not establish it or rule out risk premia
  - early out-of-sample gains are also consistent with excess shrinkage before the computer era (Fig. 6)

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# Appendix: Case Study, Replication, Limitations and Extensions

- **Replicate with the course data** (stock-level CRSP files are not part of them)
  - the 49 value-weighted industry portfolios of the main replication slides; same predictors, standardisation and windows
  - the size and price filters do not apply to portfolios
- **Limits stated by the authors**
  - lagged returns are equilibrium outcomes, not the exogenous $X$ of the model: the application illustrates the wedge, it does not test the model
  - a $t$-test on the monthly $r_{OOS}$ is not a valid test of $\mathbb{E}[r_{OOS}] = 0$: an expectation that also averages over the prior distribution of $g$
  - the backward result may rely on i.i.d. cash-flow growth, homogeneous investors and constant characteristics
- **Open extensions**
  - time series: the dividend–price regression of section 3 has a single predictor ($J = 1$), yet an in-sample $R^2$ of 0.12% and $R^2_{OS} = -0.51\%$: what explains the gap, if not the number of predictors?
  - nonlinear models (trees, neural networks Gu, Kelly & Xiu (2020)) add functional flexibility without necessarily changing $J$: does the wedge grow?
  - AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

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# Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, difference-in-differences (DiD), structural break), efficiency measure. Flag any entry you are unsure exists.`
  - Resolve every DOI on Crossref; the title and authors must match
- **Hypothesis**
  - `Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in rho1 and VR(q), and a variable that separates it from the others.`
- **Data and code**
  - `Read the daily Bitcoin closing prices from the course data; log returns; rho1 with a HAC (Newey-West) standard error of the estimate and the Lo-MacKinlay VR(5) with z*, in two windows of equal length around 11 January 2024.`

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# Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: window lengths, weekday-only closes, VR(q) for q = 2 to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).`
- **Critique**
  - `Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in rho1 and name a better outcome variable.`
- **Report**
  - Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
  - Prompts that shaped a decision go into `AI_USE.md`

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# Appendix: What the Researcher Must Check

- **i.i.d. standard errors**: $1/\sqrt{T}$ rejects too often under volatility clustering
  - test: robust SE under $H_0$ and $z^*$, as in the weak-form tests of this chapter; HAC SE for intervals and before/after comparisons
- **Unequal windows**: different power before and after
  - test: equal lengths, fixed in advance
- **Event date chosen from the data**
  - test: the launch date (first trading day) only, plus a placebo distribution at dates without an event
- **Confounding**: before/after is not causal
  - test: a control asset (Ethereum before July 2024) in a difference-in-differences
  - causal only under the conditions of the difference-in-differences slide; the common post-period (Jan–Jul 2024) is short
- **Many statistics**: $\rho_1$, VR(2–20), runs
  - test: Chow–Denning or Holm, not the smallest single $p$-value
- **Low power**: report the MDE of the Estimation and Power slide

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# Appendix: Further Study on Quantinar

## Advanced courses on quantinar.com

- Efficiency of cryptocurrency markets - A GMM-based analysis
- Generalized Method of Moments
- Market Responses to ETH Development Milestones

## Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_02
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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# Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AMH**: Adaptive Markets Hypothesis
- **AR(p)**: AutoRegressive model of order p
- **AR**: Abnormal Return, in event studies.
- **BDS**: Brock–Dechert–Scheinkman (test)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BIST**: Borsa İstanbul — *the Istanbul Stock Exchange (BIST 100 index)*
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BUX**: Budapesti Értéktőzsde index — *the Budapest Stock Exchange index*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **CAR**: Cumulative Abnormal Return
- **CD**: Chow–Denning (multiple variance-ratio test)
- **CI**: Confidence Interval
- **CME**: Chicago Mercantile Exchange
- **CRSP**: Center for Research in Security Prices
- **DFA**: Detrended Fluctuation Analysis
- **DiD**: Difference-in-Differences

## Acronyms used in this material (2/3)

- **EMH**: Efficient Market Hypothesis
- **EOD**: End of Day
- **EODHD**: EOD Historical Data (market data provider)
- **ETF**: Exchange-Traded Fund
- **fGn**: fractional Gaussian noise
- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **GARCH**: Generalized AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HK**: Hong Kong
- **IPC**: Índice de Precios y Cotizaciones — *the Mexican Stock Exchange index*
- **MDE**: Minimum Detectable Effect
- **MDH**: Martingale Difference Hypothesis
- **MIT**: Massachusetts Institute of Technology
- **ML**: Machine Learning
- **MSPE**: Mean Squared Prediction Error
- **NBER**: National Bureau of Economic Research

# Acronyms used in this material (3/3)

- **NW**: Newey–West (standard errors)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OOS**: Out-Of-Sample
- **OS**: Out-of-Sample (subscript of $R^2_{OS}$)
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **RW1**: Random Walk 1 (i.i.d. increments)
- **RW2**: Random Walk 2 (independent increments)
- **RW3**: Random Walk 3 (uncorrelated increments)
- **SD**: Standard Deviation
- **SE**: Standard Error
- **SPA**: Superior Predictive Ability (test)
- **SPDR**: Standard & Poor's Depositary Receipts
- **TOM**: Turn Of the Month
- **TSMOM**: Time-Series MOMentum
- **UK**: United Kingdom
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VR**: Variance Ratio
- **WIG20**: Warszawski Indeks Giełdowy 20 — *the Warsaw Stock Exchange index of the 20 largest companies*

# References (1/7)


- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- Anis, A.A., Lloyd, E.H. (1976). The expected value of the adjusted rescaled Hurst range of independent normal summands. *Biometrika*, 63(1), 111–116.
- Ariel, R.A. (1987). A monthly effect in stock returns. *Journal of Financial Economics*, 18(1), 161–174.
- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bartels, R. (1982). The rank version of von Neumann's ratio test for randomness. *Journal of the American Statistical Association*, 77(377), 40–46.
- Blanchard, O.J., Watson, M.W. (1982). Bubbles, rational expectations and financial markets. In P. Wachtel (ed.), *Crises in the Economic and International Financial System*, Lexington Books, 295–315.
- Boehmer, E., Musumeci, J., Poulsen, A.B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2), 253–272.
- Brock, W.A., Dechert, W.D., Scheinkman, J.A., LeBaron, B. (1996). A test for independence based on the correlation dimension. *Econometric Reviews*, 15(3), 197–235.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). The Econometrics of Financial Markets. Princeton University Press.
- Campbell, J.Y., Shiller, R.J. (1988). The dividend-price ratio and expectations of future dividends and discount factors. *Review of Financial Studies*, 1(3), 195–228.
- Campbell, J.Y., Thompson, S.B. (2008). Predicting excess stock returns out of sample: can anything beat the historical average? *Review of Financial Studies*, 21(4), 1509–1531.
- Campbell, J.Y., Yogo, M. (2006). Efficient tests of stock return predictability. *Journal of Financial Economics*, 81(1), 27–60.
- Choi, I. (1999). Testing the random walk hypothesis for real exchange rates. *Journal of Applied Econometrics*, 14(3), 293–308.
- Chordia, T., Goyal, A., Saretto, A. (2020). Anomalies and false rejections. *Review of Financial Studies*, 33(5), 2134–2179.

# References (2/7)


- Chow, K.V., Denning, K.C. (1993). A simple multiple variance ratio test. *Journal of Econometrics*, 58(3), 385–401.
- Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- Cochrane, J.H. (2008). The dog that did not bark: a defense of return predictability. *Review of Financial Studies*, 21(4), 1533–1575.
- De Bondt, W.F.M., Thaler, R. (1985). Does the stock market overreact? *Journal of Finance*, 40(3), 793–805.
- De Long, J.B., Shleifer, A., Summers, L.H., Waldmann, R.J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4), 703–738.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Escanciano, J.C., Lobato, I.N. (2009). An automatic Portmanteau test for serial correlation. *Journal of Econometrics*, 151(2), 140–149.
- Escanciano, J.C., Velasco, C. (2006). Generalized spectral tests for the martingale difference hypothesis. *Journal of Econometrics*, 134(1), 151–185.
- Fama, E.F. (1970). Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Fama, E.F. (1991). Efficient capital markets: II. *Journal of Finance*, 46(5), 1575–1617.
- Fama, E.F., Fisher, L., Jensen, M.C., Roll, R. (1969). The adjustment of stock prices to new information. *International Economic Review*, 10(1), 1–21.
- Fama, E.F., French, K.R. (1988). Dividend yields and expected stock returns. *Journal of Financial Economics*, 22(1), 3–25.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- French, K.R. (1980). Stock returns and the weekend effect. *Journal of Financial Economics*, 8(1), 55–69.
- French, K.R. (2026). Kenneth R. French Data Library. Tuck School of Business, Dartmouth College.

# References (3/7)


- FTSE Russell (2019). FTSE Russell announces results of country classification review (press release, 26 September 2019).
- Goyal, A., Welch, I., Zafirov, A. (2024). A comprehensive 2022 look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 37(11), 3490–3557.
- Grossman, S.J., Stiglitz, J.E. (1980). On the impossibility of informationally efficient markets. *American Economic Review*, 70(3), 393–408.
- Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Hansen, P.R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380.
- Harvey, C.R., Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5), 2503–2553.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Heston, S.L., Sadka, R. (2008). Seasonality in the cross-section of stock returns. *Journal of Financial Economics*, 87(2), 418–445.
- Hurst, H.E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116(1), 770–799.
- Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Kahneman, D., Tversky, A. (1979). Prospect theory: an analysis of decision under risk. *Econometrica*, 47(2), 263–291.
- Kendall, M.G. (1954). Note on bias in the estimation of autocorrelation. *Biometrika*, 41(3–4), 403–404.
- Kia, K., Liu, B., Li, Q., Song, V., Xu, K. (2026). Price discovery in Bitcoin ETF market. *Financial Review*, 61(2), 435–449.

# References (4/7)


- Kim, J.H. (2006). Wild bootstrapping variance ratio tests. *Economics Letters*, 92(1), 38–43.
- Kim, J.H., Shamsuddin, A. (2008). Are Asian stock markets efficient? Evidence from new multiple variance ratio tests. *Journal of Empirical Finance*, 15(3), 518–532.
- Kim, J.H., Shamsuddin, A., Lim, K.-P. (2011). Stock return predictability and the adaptive markets hypothesis: evidence from century-long U.S. data. *Journal of Empirical Finance*, 18(5), 868–879.
- Kolari, J.W., Pynnönen, S. (2010). Event study testing with cross-sectional correlation of abnormal returns. *Review of Financial Studies*, 23(11), 3996–4025.
- Kristoufek, L. (2018). On Bitcoin markets (in)efficiency and its evolution. *Physica A*, 503, 257–262.
- Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.
- Lakonishok, J., Smidt, S. (1988). Are seasonal anomalies real? A ninety-year perspective. *Review of Financial Studies*, 1(4), 403–425.
- Lim, K.-P., Brooks, R. (2011). The evolution of stock market efficiency over time: a survey of the empirical literature. *Journal of Economic Surveys*, 25(1), 69–108.
- Liu, Y., Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727; NBER Working Paper 24877.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (1991). Long-term memory in stock market prices. *Econometrica*, 59(5), 1279–1313.
- Lo, A.W. (2004). The adaptive markets hypothesis. *Journal of Portfolio Management*, 30(5), 15–29.
- Lo, A.W. (2017). Adaptive Markets: Financial Evolution at the Speed of Thought. Princeton University Press.
- Lo, A.W., MacKinlay, A.C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*, 1(1), 41–66.

# References (5/7)


- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- MacKinlay, A.C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13–39.
- Malkiel, B.G. (2003). The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1), 59–82.
- Martin, I.W.R., Nagel, S. (2022). Market efficiency in the age of big data. *Journal of Financial Economics*, 145(1), 154–177.
- McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Moskowitz, T.J., Ooi, Y.H., Pedersen, L.H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.
- Nadarajah, S., Chu, J. (2017). On the inefficiency of Bitcoin. *Economics Letters*, 150, 6–9.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Pele, D.T., Lin, M.-B., Wang, R., Gaman, S. (2025). Market responses to ETH development milestones. IDA Institute Digital Assets, Bucharest University of Economic Studies.
- Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- Pele, D.T., Mazurencu-Marinescu, M., Nijkamp, P. (2013). Herding behaviour, bubbles and log periodic power laws in illiquid stock markets: a case study on the Bucharest Stock Exchange. SSRN Working Paper.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets? *European Journal of Finance*, 29(10), 1064–1105.
- Peng, C.-K., Buldyrev, S.V., Havlin, S., Simons, M., Stanley, H.E., Goldberger, A.L. (1994). Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2), 1685–1689.

# References (6/7)


- Roberts, H.V. (1959). Stock-market "patterns" and financial analysis: methodological suggestions. *Journal of Finance*, 14(1), 1–10.
- Robinson, P.M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5), 1630–1661.
- Romano, J.P., Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282.
- Rozeff, M.S., Kinney, W.R. (1976). Capital market seasonality: the case of stock returns. *Journal of Financial Economics*, 3(4), 379–402.
- Samuelson, P.A. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review*, 6(2), 41–49; reprinted 2015, World Scientific.
- Shiller, R.J. (1981). Do stock prices move too much to be justified by subsequent changes in dividends? *American Economic Review*, 71(3), 421–436.
- Shiller, R.J. (2026). U.S. stock markets 1871–present and CAPE ratio (monthly data, accessed September 2026).
- Shleifer, A., Vishny, R.W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1), 35–55.
- Stambaugh, R.F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375–421.
- Sullivan, R., Timmermann, A., White, H. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance*, 54(5), 1647–1691.
- Summers, L.H. (1986). Does the stock market rationally reflect fundamental values? *Journal of Finance*, 41(3), 591–601.
- Tak, R., Pele, D.T. (2025). LLM-driven stock prediction: capturing market trends with LLaMA. *Proceedings of the International Conference on Business Excellence*, 19(1), 529–540.
- Tak, R., Schlamp, S., Osterrieder, J., Pele, D.T. (2026). Racing the news: nanosecond market efficiency at Eurex and CME. SSRN Working Paper.

# References (7/7)


- Tirole, J. (1985). Asset bubbles and overlapping generations. *Econometrica*, 53(6), 1499–1528.
- Tran, V.L., Leirvik, T. (2020). Efficiency in the markets of crypto-currencies. *Finance Research Letters*, 35, 101382.
- Urquhart, A. (2016). The inefficiency of Bitcoin. *Economics Letters*, 148, 80–82.
- Urquhart, A., McGroarty, F. (2016). Are stock markets really efficient? Evidence of the adaptive market hypothesis. *International Review of Financial Analysis*, 47, 39–49.
- Wald, A., Wolfowitz, J. (1940). On a test whether two samples are from the same population. *Annals of Mathematical Statistics*, 11(2), 147–162.
- Welch, I., Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.